**MECH 325**

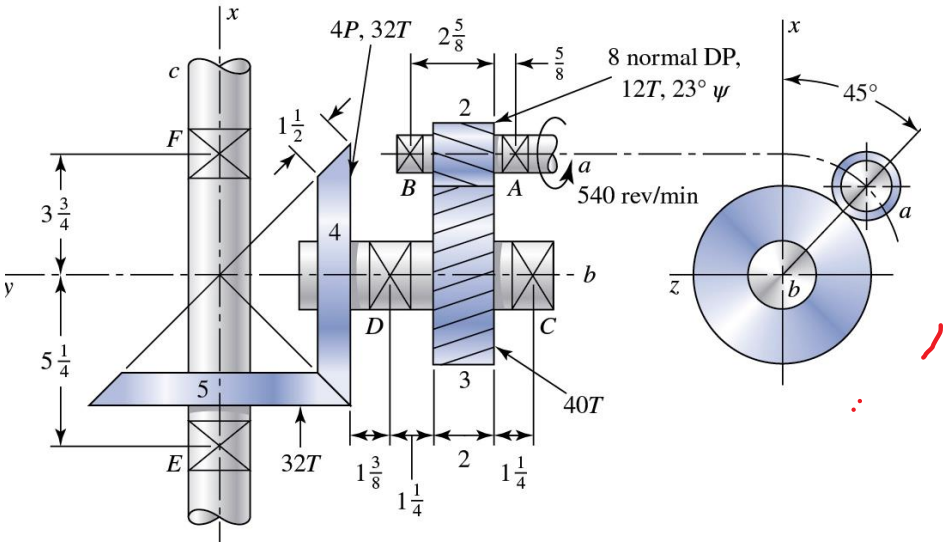
**Homework Assignment #2**

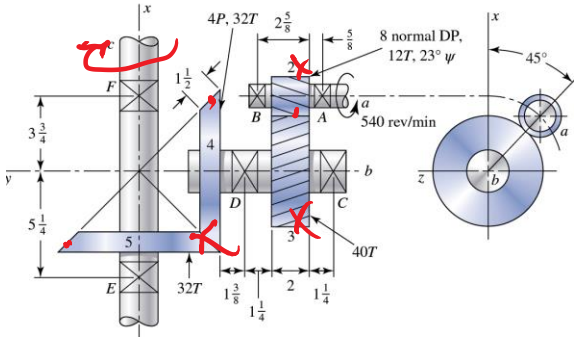
**Due Oct. 21**

**Problem 1**

Shigley 13-19. The mechanism train consist of a pair of helical gears and a pair of mitre gears. The helical gears have a 17.5° normal angle and a helix angle of 23°. Find:

a) The speed and direction of shaft c. Assume shaft a is turning CCW.
b) The distance between shafts a and b (assume teeth are meshed properly)
c) The pitch diameter of the miter gears.



**13-19 (a)**

$$n_c = \frac{12}{40}\cdot\frac{1}{1}(540) = 162 \text{ rev/min}$$

cw about $x$, as viewed from the positive $x$ axis

*Ans.*

**(b)** $d_P = 12/(8\cos 23^\circ) = 1.630$ in

$d_G = 40/(8\cos 23^\circ) = 5.432$ in

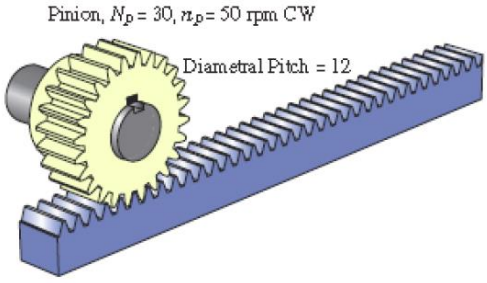
$$\frac{d_P + d_G}{2} = 3.531 \text{ in} \quad \textit{Ans.}$$

**(c)** $$d = \frac{N}{P} = \frac{32}{4} = 8 \text{ in} \quad \textit{Ans.}$$

**Problem 2**

Mott 8-81. A rack is driven by a pinion that has a rotational speed of 50 rpm. The pinion has 30 teeth with a diametrical pitch of 12 teeth/inch. The rack has a length of 7 feet (See Table 8-10 for rack specifications). Determine the following:

a) The pitch diameter of the pinion
b) The distance from the pitch line to the back of the rack
c) The distance from the pinion center to the back of the rack
d) The linear velocity of the rack
e) The time required to move the rack 7.0 feet.
f) The number of revolutions required to move the rack 7.0 feet



GIVEN: $n_p = 50$ rpm

$N_p = 30$ teeth

$P_d = 12$ teeth/in

Mott, Table 8-10

TABLE 8–10 Example rack specifications

| Diametral pitch | Pitch line to back ($B$) | Overall thickness | Face width | Nominal length [ft] |
|---|---|---|---|---|
| 64 | 0.109 | 0.125 | 0.125 | 2 |
| 48 | 0.104 | 0.125 | 0.125 | 2 |
| 32 | 0.156 | 0.187 | 0.187 | 4 |
| 24 | 0.208 | 0.250 | 0.25 | 4 |
| 20 | 0.450 | 0.500 | 0.5 | 6 |
| 16 | 0.688 | 0.750 | 0.75 | 6 |
| 12 | 0.917 | 1.000 | 1 | 6 |
| 10 | 1.150 | 1.250 | 1.25 | 6 |
| 8 | 1.375 | 1.500 | 1.5 | 6 |
| 6 | 1.333 | 1.500 | 2 | 6 |
| 5 | 1.300 | 1.500 | 2.5 | 6 |
| 4 | 1.750 | 2.000 | 3.5 | 6 |

1. $d_p = \frac{N_p}{P_d} = \frac{30 \text{ teeth}}{12 \text{ teeth/in}}$

$\boxed{d_p = 2.5 \text{ in}}$

2. $\boxed{B = 0.917 \text{ in}}$ [Table 8-10, row 7]

3. $\text{dist.} = \frac{d_p}{2} + B = \frac{2.5}{2} + 0.917 = 2.167 \text{ in}$

$\boxed{\text{distance from pinion center to back of rack} = 2.167 \text{ in}}$

4. $V_{rack} = r \cdot \omega$

$= \frac{d_p}{2} \cdot n_p \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = \frac{2.5 \text{ in}}{2} \cdot \left(50 \frac{\text{rev}}{\text{min}}\right) \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 32.725 \text{ ft/min}$

$\boxed{V_{rack} = 32.725 \text{ ft/min}}$

5. $v = \frac{s}{t} \rightarrow t = \frac{s}{v} = \frac{7.0 \text{ ft}}{32.725 \text{ fpm}} = 0.2139 \text{ min} \times \frac{60 \text{ s}}{\text{min}} = 12.834 \text{ s}$

$\boxed{t = 12.834 \text{ s}}$

6. $C_p = \pi \cdot P_d = 7.8539 \text{ in}$ (pitch circumference)

$N_{rev} = \frac{7.0 \text{ ft} \cdot 12 \text{ in/ft}}{7.8539} = 10.695 \text{ revolutions}$

$\boxed{N_{rev} = 10.695 \text{ revolutions}}$

OR

$s_{RACK} = \frac{D_p}{2} \cdot \theta_p$ [Eq. 8-37]

$\theta_p = \frac{2\left(7 \text{ ft} \times \frac{12 \text{ in}}{\text{ft}}\right)}{2.5 \text{ in}} = 67.2 \text{ rad}$

SAME ANSWER ✓

$67.2 \text{ rad} \times 1 \text{ rev}/2\pi \text{ rad} = 10.695 \text{ rev}$

**Problem 3**

Mott 8-42. A helical gear has a normal diametrical pitch of 12, a normal pressure angle of 20°, 48 teeth, a face-width of 1.50 inches and a helix angle of 45°. Compute the circular pitch, axial pitch, pitch diameter, and transverse pressure angle. Compute the number of axial pitches in the face width. Does the *face contact ratio* meet the requirement?

GIVEN: $P_{nd} = 12$ teeth/in

$\phi_n = 20°$ // $\psi = 45°$

$N = 48$ Teeth // $F = 1.50$ in

CIRCULAR PITCH (TRANSVERSE), $P_t$

$P_t = \pi D/N = \pi/P_d$ [Eq. 8-12] $\quad P_t = \frac{\pi}{8.4853}$ $\quad \boxed{P_t = 0.370 \text{ in}}$

$P_{nd} = P_d/\cos\psi$ [Eq. 8-16]

↳ $P_d = P_{nd} \cdot \cos\psi = 12 \frac{\text{teeth}}{\text{in}} \cdot \cos 45° \quad \underline{P_d = 8.4853 \text{ teeth/in}}$

NORMAL CIRCULAR PITCH, $P_n$

$P_n = P_t \cos\psi$ [Eq. 8-13] $= 0.37024 \cos 45° \quad \boxed{P_n = 0.262 \text{ in}}$

AXIAL PITCH

$P_x = P_t/\tan\psi$ [Eq. 8-14]

$= 0.370/\tan(45°) = 0.370$

NOTE: $P_x = P_t$ for $\psi = 45°$ ($\tan 45° = 1$)

$\boxed{P_x = 0.370 \text{ in}}$

PITCH DIAMETER

$P_d = \frac{N}{D}$ [Eq. 8-15]

$P_{nd} = P_d/\cos\psi$ [Eq. 8-16]

Solve for D:

$D = \frac{N}{P_d} = \frac{N}{P_{nd} \cdot \cos 45°} = \frac{48 \text{ teeth}}{12 \frac{\text{teeth}}{\text{in}} \cdot \cos 45°} = 5.657$

$\boxed{D = 5.657 \text{ in}}$

TRANSVERSE PRESSURE ANGLE

$\tan\phi_n = \tan\phi_t \cos\psi$ [Eq. 8-11]

$\phi_t = \tan^{-1}\left(\frac{\tan\phi_n}{\cos\psi}\right) = \tan^{-1}\left(\frac{\tan 20°}{\cos 45°}\right) \quad \boxed{\phi_t = 27.24°}$

# of AXIAL PITCHES IN FACE WIDTH

FACE CONTACT RATIO $= F/P_x = \frac{1.50 \text{ in}}{0.370 \text{ in/pitch}} = \boxed{4.054 \text{ pitches}}$

[See pg. 325]

↳ > 2.0, so full helical action occurs

**Problem 4**

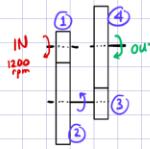
**Shigley 13-43 (modified)**

A speed-reducer gearbox containing a compound reverted gear train (an example figure is shown on Shigley 13-39). The gearbox transmits 34 HP with an input speed of 1150 rpm. Spur gears with 20° pressure angle are used with 17 teeth on each of the small teeth and 48 teeth on each of the larger gears. A diametrical pitch of 10 teeth/inch is proposed.

a) Determine the speeds of the intermediate and the output shafts.
b) Determine the pitch line velocities for each of the gears.
c) Determine the magnitude of the tangential and radial forces transmitted between each set of gears.
d) Determine the input torque
e) Determine the output torque neglecting frictional losses

$H = 30$ hp
$n_{in} = 1200$ rpm
$N_p = 18$ // $N_G = 54$ // $\phi = 20°$ // $P_d = 10$ teeth/in

a)




INTERMEDIATE: $n_{int} = 1200 \cdot \frac{18}{54} = 400$ → $n_{int} = 400$ rpm (opposite to input)

OUTPUT: $n_{out} = 1200 \cdot \frac{18}{54} \cdot \frac{18}{54} = 133.33$ → $n_{out} = 133.33$ rpm (same as input direction)

$e = \frac{18}{54} \cdot \frac{18}{54} = 1/9$

b) $P = \frac{N}{d_p} \rightarrow d_p = \frac{N_p}{P} = \frac{18 \text{ teeth}}{10 \text{ teeth/in}}$ $d_p = 1.8$ in

$d_G = \frac{N_G}{P} = \frac{54 \text{ teeth}}{10 \text{ teeth/in}}$ $d_G = 5.4$ in

- same pitch-line velocities for gears in mesh: $V_1 = V_2$, $V_3 = V_4$

$V = \pi d n / 12$ [Eq. 13-34] units feet/min.

$V_1 = V_2 = \frac{1.8 \text{ in}}{2} \cdot 1200 \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} =$

$= \pi(1.8)(1200)/12 = 565.487$

$V_1 = V_2 = 565.49$ ft/min.

$V_3 = V_4 = \frac{1.8 \text{ in}}{2} \cdot 400 \text{ rpm} \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}}$

$= \pi(1.8)(400)/12 = 188.496$

$V_3 = V_4 = 188.5$ ft/min.

(check: $V_3 = V_1 \cdot \frac{18}{54}$ ✓)

c) FORCES - TANGENTIAL

$T_{in} = \frac{63025 \, H}{n} = F_{1t} \cdot \frac{d_1}{2} \rightarrow F_{1t} = \frac{63025 \, (30 \text{ hp})}{1200 \text{ rpm}} \cdot \frac{2}{1.8 \text{ in}} = 1750.7$ lbs

$T_{intermediate} = F_{1t} \cdot \frac{d_2}{2} = 1750.7 \text{ lb} \cdot \frac{5.4 \text{ in}}{2} = 4726.875$ lb-in on intermediate shaft

$F_{3,t} = \frac{T_{int.}}{r_3} = \frac{T_{int}}{d_3/2} = \frac{4726.875 \text{ lb-in}}{1.8/2} = 5252.08$ lbf

TANGENTIAL

$F_{t,1-2} = 1750.7$ lbf

$F_{t,3-4} = 5252.1$ lbf

f)

FORCES - RADIAL




$F_t = F \cos\phi$ // $F = \frac{F_t}{\cos\phi}$

$F_r = F \sin\phi$

$F_r = \frac{F_t}{\cos\phi} \cdot \sin\phi = F_t \tan\phi$

$F_{r_{1-2}} = 1750.7 \cdot (\tan 20°) = 637.203$ lbf

$F_{r_{3-4}} = 5252.1 \cdot (\tan 20°) = 1911.61$ lbf

RADIAL

$F_{r,1-2} = 637.2$ lbf

$F_{r,3-4} = 1911.6$ lbf

d) INPUT TORQUE

$T_{in} = \frac{63025 \cdot H}{n_1} = \frac{63025 \cdot 30 \text{ hp}}{1200 \text{ rpm}}$

$T_{in} = 1575.625$ lb·in → $T_{IN} = 1575.6$ lb·in

e) OUTPUT TORQUE

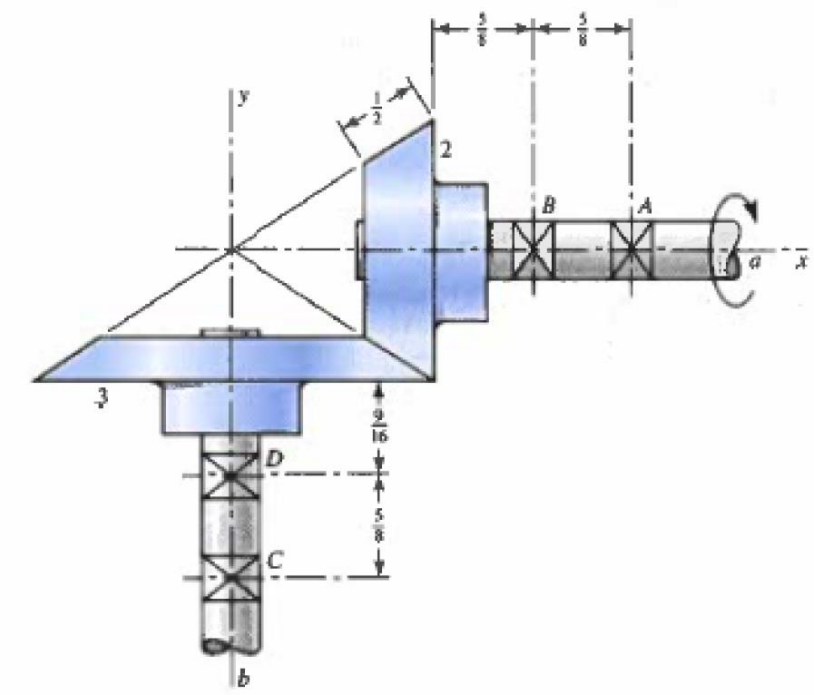
$T_{out} = T_{in} \cdot \frac{54}{18} \cdot \frac{54}{18} = 14180.625$ lb·in

$T_{OUT} = 14180.6$ lb·in

(assuming no frictional losses)

**Problem 5**

**13–50** The figure shows a 10 diametral pitch 18-tooth 20° straight bevel pinion driving a 30-tooth gear. The transmitted load is 25 lbf. Find the bearing reactions at $C$ and $D$ on the output shaft if $D$ is to take both radial and thrust loads.




Note: This solution assumes that the contact point is at the top end of the gears as noted on the figure.

$d_2 = 18/10 = 1.8$ in, $d_3 = 30/10 = 3.0$ in

$$\gamma = \tan^{-1}\left(\frac{d_2/2}{d_3/2}\right) = \tan^{-1}\left(\frac{0.9}{1.5}\right) = 30.96^\circ$$

$$\Gamma = 180^\circ - \gamma = 59.04^\circ$$

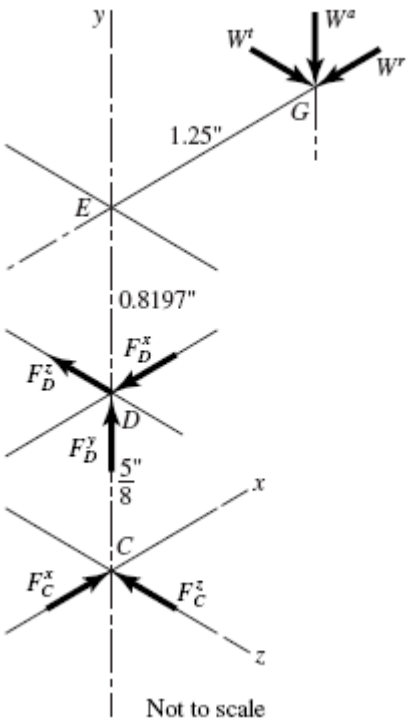
$$DE = \frac{9}{16} + 0.5\cos 59.04^\circ = 0.8197 \text{ in}$$

$$W^t = 25 \text{ lbf}$$

$$W^r = 25\tan 20^\circ \cos 59.04^\circ = 4.681 \text{ lbf}$$

$$W^a = 25\tan 20^\circ \sin 59.04^\circ = 7.803 \text{ lbf}$$

**W** = –4.681**i** – 7.803**j** +25**k**



$\mathbf{R}_{DG} = 0.8197\mathbf{j} + 1.25\mathbf{i}$

$\mathbf{R}_{DC} = -0.625\mathbf{j}$

$$\Sigma \mathbf{M}_D = \mathbf{R}_{DG} \times \mathbf{W} + \mathbf{R}_{DC} \times \mathbf{F}_C + \mathbf{T} = \mathbf{0}$$

$$\mathbf{R}_{DG} \times \mathbf{W} = 20.49\mathbf{i} - 31.25\mathbf{j} - 5.917\mathbf{k}$$

$$\mathbf{R}_{DC} \times \mathbf{F}_C = -0.625F_C^z\mathbf{i} + 0.625F_C^x\mathbf{k}$$

$$\left(20.49\mathbf{i} - 31.25\mathbf{j} - 5.917\mathbf{k}\right) + \left(-0.625F_C^z\mathbf{i} + 0.625F_C^x\mathbf{k}\right) + T\mathbf{j} = \mathbf{0}$$

$T = 31.25 \text{ lbf} \cdot \text{in}$ *Ans.*

$\mathbf{F}_C = 9.47\mathbf{i} + 32.8\mathbf{k} \text{ lbf}$ *Ans.*

$F_C = \left(9.47^2 + 32.8^2\right)^{1/2} = 34.1 \text{ lbf}$ *Ans.*

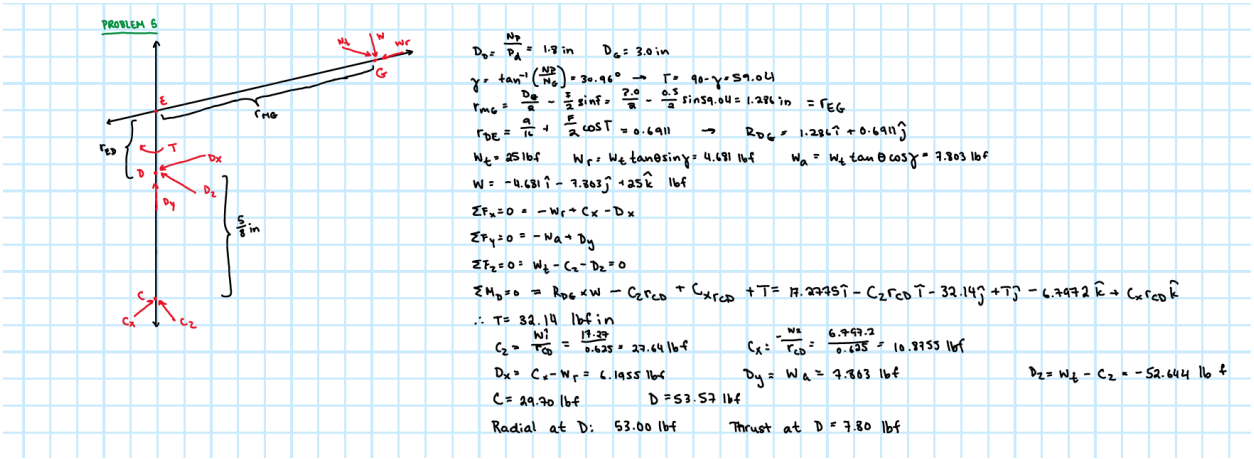
$\Sigma \mathbf{F} = 0 \qquad \mathbf{F}_D = -4.79\mathbf{i} + 7.80\mathbf{j} - 57.8\mathbf{k} \text{ lbf}$

$F_D(\text{radial}) = \left[\left(-4.79\right)^2 + \left(-57.8\right)^2\right]^{1/2} = 58.0 \text{ lbf}$ *Ans.*

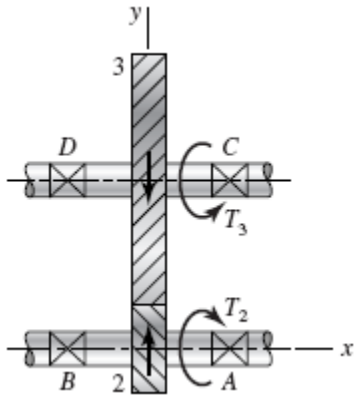
$F_D(\text{thrust}) = W^a = 7.80 \text{ lbf}$ *Ans.*

---

**Note: This solution is for the contact point at the midpoint of the gear mesh which is correct.**



**Problem 6**




Input Shaft a

**13–54** In the figure for Problem 13–35, pinion 2 is to be a right-hand helical gear having a helix angle of 30°, a normal pressure angle of 20°, 16 teeth, and a normal diametral pitch of 6 teeth/in. A motor delivers 25 hp to shaft *a* at a speed of 1720 rev/min clockwise about the *x* axis. Gear 3 has 42 teeth. Find the reaction exerted by bearings *C* and *D* on shaft *b*. One of these bearings is to take both radial and thrust loads. This bearing should be selected so as to place the shaft in compression.

**13-54**

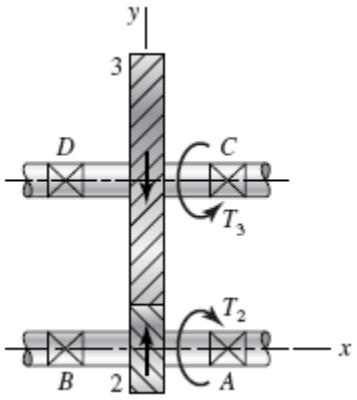
$$P_t = 6\cos 30^\circ = 5.196 \text{ teeth/in}$$

$$d_3 = \frac{42}{5.196} = 8.083 \text{ in}$$

$$\phi_t = 22.8^\circ$$

$$d_2 = \frac{16}{5.196} = 3.079 \text{ in}$$

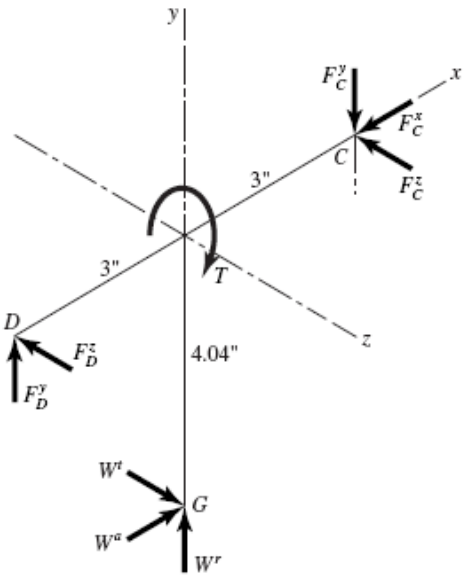
$$T_2 = \frac{63025(25)}{1720} = 916 \text{ lbf} \cdot \text{in}$$



$$W^t = \frac{T}{r} = \frac{916}{3.079/2} = 595 \text{ lbf}$$

$$W^a = 595 \tan 30^\circ = 344 \text{ lbf}$$

$$W^r = 595 \tan 22.8^\circ = 250 \text{ lbf}$$




$$\mathbf{W} = 344\mathbf{i} + 250\mathbf{j} + 595\mathbf{k} \text{ lbf}$$

$$\mathbf{R}_{DC} = 6\mathbf{i}, \quad \mathbf{R}_{DG} = 3\mathbf{i} - 4.04\mathbf{j}$$

$$\Sigma \mathbf{M}_D = \mathbf{R}_{DC} \times \mathbf{F}_C + \mathbf{R}_{DG} \times \mathbf{W} + \mathbf{T} = \mathbf{0} \quad (1)$$

$$\mathbf{R}_{DG} \times \mathbf{W} = -2404\mathbf{i} - 1785\mathbf{j} + 2140\mathbf{k}$$

$$\mathbf{R}_{DC} \times \mathbf{F}_C = -6F_C^z\mathbf{j} + 6F_C^y\mathbf{k}$$

Substituting and solving Eq. (1) gives

$$\mathbf{T} = 2404\mathbf{i} \text{ lbf} \cdot \text{in}$$

$$F_C^z = -297.5 \text{ lbf}$$

$$F_C^y = -365.7 \text{ lbf}$$

$$\Sigma \mathbf{F} = \mathbf{F}_D + \mathbf{F}_C + \mathbf{W} = \mathbf{0}$$

Substituting and solving gives

$$F_C^x = -344 \text{ lbf}$$

$$F_D^y = 106.7 \text{ lbf}$$

$$F_D^z = -297.5 \text{ lbf}$$

$$\mathbf{F}_C = -344\mathbf{i} - 356.7\mathbf{j} - 297.5\mathbf{k} \text{ lbf} \quad \textit{Ans.}$$

$$\mathbf{F}_D = 106.7\mathbf{j} - 297.5\mathbf{k} \text{ lbf} \quad \textit{Ans.}$$

**Problem 7. Mott 9-60 (modified)**

A pair of spur gears is to be designed for a reciprocating compressor transmission. The machine requires 5 HP with the pinion speed turning at 1200 rpm (exact) with the output gear speed to be between 385 and 390 rpm. An electric motor is driving the pinion. The recommended life for the machine is 15000 hours of operation. Assume commercial gear quality ($A_v$ = 10, $Q_v$ = 7).

Design a set of spur gears specifying:

a) The diametrical pitch (use standard values)
b) The number of teeth in each gear (be sure to avoid interference)
c) The pitch diameters of each gear and the center distance
d) The face width
e) The transverse and radial forces
f) Determine the bending and contact stress numbers for the pinion only. Be sure to list any assumptions in this analysis.
g) Determine a suitable material and hardness for the pinion (you can assume the same material will be selected for the gear). Specify the alloy and heat treatment.
h) What is the final safety factor for the design (pinion only)?

(See Mott, Pg 403-405)

$H = 5.0$ hp

$n = 1200$ rpm

$n_G \approx 385 - 390$ rpm

From Table 9-1:

Source: Electric Motor
↳ Uniform

Driven: Reciprocating Compressor
↳ Moderate Shock

Overload Factor:

$K_o = 1.50$

$P_{des} = P_{nom} \cdot K_o = 5.0 \text{ hp} \cdot 1.5 = 7.5 \text{ hp}$ $\quad P_{des} = 7.5$ hp

Figure 9-11: @ $P_{des} = 7.5$ hp, $n = 1200$ rpm

→ choose $P_d = 10$ T/in

Choose $N_p = 18$ teeth

Nominal $VR = \frac{n_p}{n_G} = \frac{1200}{387.5} \approx 3.098$

Approx $N_G = 18 \cdot 3.098 \rightarrow N_G \approx 55.742$ round to 56 $\quad N_G = 56$

Actual $VR = \frac{56}{18} = 3.111$

Actual output speed: $n_G = 1200 \cdot \frac{18}{56} = 385.7$ rpm ✓ OK

$D_p = \frac{N_p}{P_d} = \frac{18}{10} = 1.8$ in

$D_G = \frac{N_G}{P_d} = \frac{56}{10} = 5.6$ in

$CD = \frac{D_p}{2} + \frac{D_G}{2} = \frac{1.8 + 5.6}{2}$

$CD = 3.7$ in

Face Width: use rule of thumb $F = \frac{12}{P_d}$ $\quad F = 1.2$ in

$V_t = \frac{\pi D_p n_p}{12} = \frac{\pi}{12}(1.8 \text{ in})(1200 \text{ rpm})$ $\quad V_t = 565.49$ ft/min

Transmitted Load: $W_t = \frac{33000\,(P)}{V_t} = \frac{33000\,(5.0 \text{ hp})}{565.49 \text{ ft/min}} = 291.78$ lb

$W_t = 291.8$ lb

Material: Choose Steel gears: $C_p = 2300$ [Table 9-7]

Quality: $A_v = 10$ [Table 9-5]

From Fig. 9-16: $K_v = 1.26$

Fig. 9-10: Choose 20° pressure angle, full depth teeth

$J_p = 0.32$

$J_G = 0.40$ } BENDING GEOMETRY FACTORS

Fig. 9-17: $I = 0.102$ ] PITTING GEOMETRY FACTOR

Eq. 9-17 / Fig. 9-12 / 9-13

$$C_{pf} = \frac{F}{10D_p} - 0.0375 + 0.0125F = \frac{1.2}{10(1.8)} - 0.0375 + 0.0125(1.2)$$

$C_{pf} = 0.04417$ PINION PROPORTION

$$\frac{F}{D_p} = \frac{1.2}{1.8} = 0.667$$

$$C_{ma} = 0.127 + 0.0158F - 1.093 \times 10^{-4} F^2$$ $C_{ma} = 0.1458$ MESH ALIGNMENT

$K_m = 1.0 + C_{pf} + C_{ma}$ $K_m = 1.19$ LOAD-DISTRIBUTION FACTOR

$K_s = 1.00$ [Table 9.2] SIZE FACTOR

$K_B = 1.00$ (solid gear blank) Fig. 9-14 RIM THICKNESS FACTOR

$SF = 1.0$ SERVICE FACTOR

$K_R = 1.00$ RELIABILITY (0.99)

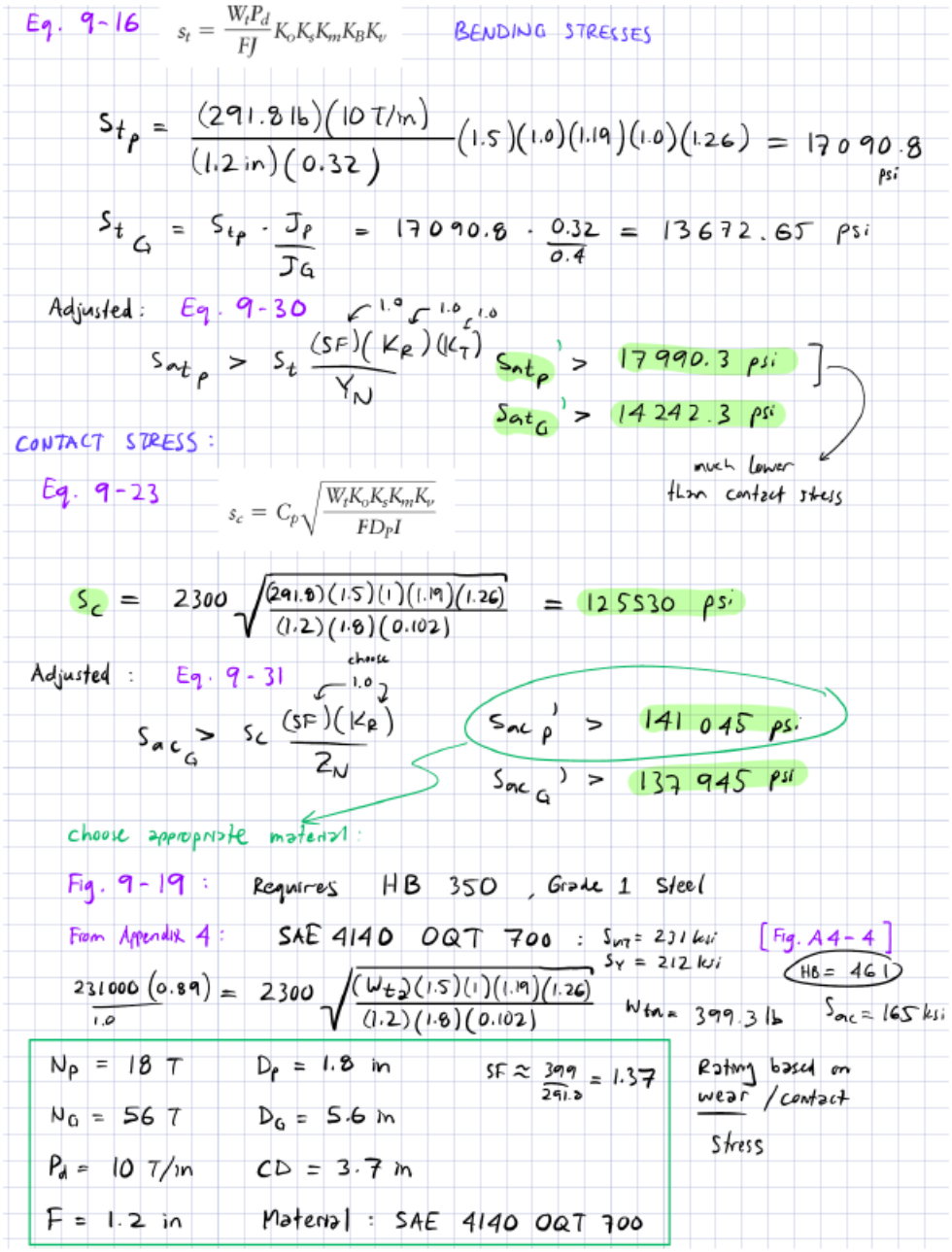
Design Life = 15000 hours $N_C = 60(L)(n)(q)$ Eq. 9-27

$N_{CP} = 60(15000)(1200)(1) = 1.08 \times 10^9$ cycles

$N_{CG} = 60(15000)(385.7)(1) = 3.471 \times 10^8$ cycles

Fig. 9-21 $Y_{N_p} = 0.95$ $Y_{N_G} = 0.96$ BENDING

Fig. 9-22 $Z_{NP} = 0.89$ $Z_{N_G} = 0.91$ PITTING




**Problem 8. Mott 10-5 (modified)**

Replace the gear set from Problem 7 with a pair of helical gears having a 25° helix angle and 20° transverse pressure angle. To simplify the question use your selected pitch and speed ratio from Problem 7 (diametrical pitch and numbers of teeth).

Design a set of spur gears specifying:

a) The pitch diameters of each gear and the center distance
b) The face width (note face contact ratio requirements)

c) The transverse, radial, and axial loads
d) Determine the bending and contact stress numbers for the pinion only. Be sure to list any assumptions in this analysis.
e) Determine a suitable material and hardness for the pinion (you can assume the same material will be selected for the gear). Specify the alloy and heat treatment.
f) What is the final safety factor for the design (pinion only)?
g) How does this result compare with Problem 7

*Problem 10-5*

| ***nitial Input Data:*** | | | |
|---|---|---|---|
| Input Power: | $P$ = | 5 | hp |
| Input Speed: | $n_P$ = | 1200 | rpm |
| ***erse Diametral Pitch, $P_d$:*** | $P_d$ = | 18 | |
| Number of Pinion Teeth: | $N_P$ = | 18 | |
| Desired Output Speed: | $n_G$ = | 387.5 | rpm |
| ***uted number of gear teeth:*** | | 55.7 | |
| ***Chosen No. of Gear Teeth:*** | $N_G$ = | 56 | |
| ***Computed data:*** | | | |
| Actual Output Speed: | $n_G$ = | 385.7 | rpm |
| Gear Ratio: | $m_G$ = | 3.11 | |
| Pitch Diameter - Pinion: | $D_P$ = | 1.000 | in |
| Pitch Diameter - Gear: | $D_G$ = | 3.111 | in |
| Center Distance: | $C$ = | 2.056 | in |
| Pitch Line Speed: | $v_t$ = | 314 | ft/min |
| Transmitted Load: | $W_t$ = | 525 | lb |

| ***Secondary Input Data:*** | | | | |
|---|---|---|---|---|
| ***Transverse pressure angle:*** | $\phi_t$ = | 20.00 | deg | |
| Helix angle: | $\psi$ = | 25.0 | deg | |
| Axial Pitch: | $p_x$ = | 0.374 | in | |
| ***ce Width (2 x Axial Pitch):*** | $F_{min}$ = | 0.749 | in | |
| *Enter:* Face Width: | $F$ = | 1.200 | in | |
| *Enter:* Elastic Coefficient: | $Cp$ = | 2300 | | Table 9-7 |
| *Enter:* Quality Number: | $A_v$ = | 11 | | Table 9-5 |
| REF: $N_P$, $N_G$ = | 18 | 56 | | |
| ***ending Geometry Factors:*** | | | | |
| Pinion: | $J_P$ = | 0.453 | | Fig 10-5,6,7 |
| Gear: | $J_G$ = | 0.486 | | Fig 10-5,6,7 |
| ***:Pitting Geometry Factor:*** | $I$ = | 0.205 | | Tab. 10-1,2 |
| REF: $m_G$ = | 3.11 | | | |
| Axial Force: | $W_x$ = | 245 | lb | |
| Radial Force: | $W_r$ = | 191 | lb | |

| ***Factors in Design Analysis:*** | | | | |
|---|---|---|---|---|
| Alignment Factor, $K_m$=1.0+$C_{pf}$+$C_{ma}$ | If F<1.0 | If F>1.0 | F/$D_P$ = 1.20 | |
| Pinion Proportion Factor, $C_{pf}$ = | 0.095 | 0.098 | [0.50 < F/$D_P$ < 2.00] | |
| *Enter:* $C_{pf}$ = | 0.098 | Figure 9-12 | | |
| Type of gearing: | Open | Commer. | Precision | Ex. Prec. |
| Mesh Alignment Factor, $C_{ma}$ = | 0.267 | 0.146 | 0.083 | 0.050 |
| *Enter:* $C_{ma}$ = | 0.146 | Figure 9-13 | | |
| Alignment Factor: $K_m$ = | 1.24 | [Computed] | | |
| Overload Factor: $K_o$ = | 1.50 | Table 9-1 | | |
| Size Factor: $K_s$ = | 1.00 | Table 9-2: Use 1.00 if $P_d$ >= 5 | | |
| Pinion Rim Thickness Factor: $K_{BP}$ = | 1.00 | Fig. 9-14: Use 1.00 if solid blank | | |
| Gear Rim Thickness Factor: $K_{BG}$ = | 1.00 | Fig. 9-14: Use 1.00 if solid blank | | |
| Dynamic Factor: $K_v$ = | 1.24 | [Computed: See Fig. 9-16] | | |
| Service Factor: $SF$ = | 1.00 | Use 1.00 if no unusual conditions | | |
| Reliability Factor: $K_R$ = | 1.00 | Table 9-11: Use 1.00 for $R$ = .99 | | |
| *Enter: Design Life:* | 15000 | hours | See Table 9-12 | |
| Pinion - Number of load cycles: $N_P$ = | 1.1E+09 | Guidelines: $Y_N$, $Z_N$ | | |
| Gear - Number of load cycles: $N_G$ = | 3.5E+08 | $10^7$ cycles | >$10^7$ | <$10^7$ |
| Bending Stress Cycle Factor: $Y_{NP}$ = | 0.94 | 1.00 | 0.94 | Fig. 9-21 |
| Bending Stress Cycle Factor: $Y_{NG}$ = | 0.96 | 1.00 | 0.96 | Fig. 9-21 |
| Pitting Stress Cycle Factor: $Z_{NP}$ = | 0.90 | 1.00 | 0.90 | Fig. 9-22 |
| Pitting Stress Cycle Factor: $Z_{NG}$ = | 0.92 | 1.00 | 0.92 | Fig. 9-22 |

***Stress Analysis: Bending***

| | | |
|---|---|---|
| Pinion: Required $s_{at}$ = | 42,786 psi | See Fig. 9-18 or |
| Gear: Required $s_{at}$ = | 39,050 psi | Table 9-9 |

***Stress Analysis: Pitting***

| | | |
|---|---|---|
| Pinion: Required $s_{ac}$ = | 179,572 psi | See Fig. 9-19 or |
| Gear: Required $s_{ac}$ = | 175,668 psi | Table 9-9 |

***Specify materials, alloy and heat treatment, for most severe requirement.***

***One possible material specification:*** Steel pinion and gear. Carburized, Grade 1

Pinion requires HRC 58 min.: SAE 4118 DOQT 300; HRC 62; Carburized

Gear requires HRC 58 min.: SAE 4118 DOQT 300; HRC 62; Carburized

***Computed stresses without modifying factors for reliability and life:***

| | | |
|---|---|---|
| Computed bending stress no. = $s_t$ = | 40,219 psi | Pinion |
| Computed bending stress no. = $s_t$ = | 37,488 psi | Gear |
| Computed contact stress no. = $s_c$ = | 161,614 psi | Pinion |
| Computed contact stress no. = $s_c$ = | 161,614 psi | Gear |

For K
B = 0.8
C = 59
Throu
Gra
HB 38
HB 34
HB 46
HB 45
Red: HB
for throug

## Similar problem but from last year

**Problem 8. (Matt 10-6 (modified))**

Replace the gear set from Problem 7 with a pair of helical gears having a 15° helix angle and 20° transverse pressure angle. To simplify the question use your selected pitch and speed ratio from Problem 7 (diametrical pitch and numbers of teeth).

Design a set of helical spur gears specifying:

1. The pitch diameters of each gear and the center distance
2. The face width (note face contact ratio requirements)
3. The transverse, radial, and axial loads
4. Determine the bending and contact stress numbers for the pinion only. Be sure to list any assumptions in this analysis.
5. Determine a suitable material and hardness for the pinion (you can assume the same material will be selected for the gear). Specify the alloy and heat treatment.
6. What is the final safety factor for the design (pinion only)?
7. How does this result compare with Problem 7

GIVEN:

$\psi = 15°$

$\phi_t = 20°$

Given in Question 7:

$P = 20$ hp

$n_p = 520$ rpm

$180 < n_G < 190$

Recommended life = 20000 hr

$A_N = 9$, $G_V = 8$

8. Find actual Output rotational (speed ($n_{G,a}$) using $VR_{actual}$

$VR_{actual} = \frac{n_p}{n_{G,a}} \longrightarrow n_{G,a} = \frac{n_p}{VR_{actual}} = \frac{520}{2.8333} = 183.532$ rpm

$n_{G,a} = 183.532$ rpm

9. Check if $n_{G,a}$ is within given range ($180 < n_G < 190$)

YES → Proceed

NO → Try new $N_p$

$180 < n_G < 190$

∴ can proceed.

10. Find the diameters of the gear and pinion ($D_G$ and $D_p$)

$D_G = \frac{N_G}{P_d} = \frac{52}{5} = 10.2$ in

$D_p = \frac{N_p}{P_d} = \frac{18}{5} = 3.6$ in

∴ $D_G = 10.2$ in, $D_p = 3.6$ in

11. Find Centre distance (C), pitch line speed ($V_t$), transmitted load ($W_t$), radial load ($W_r$), and Normal load ($W_n$)

$C = \frac{N_p + N_G}{2P_d} = \frac{D_G + D_p}{2(5)} = \frac{10.2 + 3.6}{2(5)} = 1.38$ in

$V_t = \frac{\pi D_p n_p}{12} = \frac{\pi D_G n_G}{12} = \frac{\pi(10.2)(183.532)}{12} = 490.095$ ft/min

Note: $n_G = n_{G,actual}$

$W_t = \frac{126000P}{nD} = \frac{33000P}{V_t} = \frac{33000(20)}{490.093} = 1346.68$ lb

Note: P used was Pdes

$W_r = W_t \tan(\phi) = (1346.68)\tan(20) = 490.151$ lb

$W_r = 490.151$ lb

$W_n = \sqrt{W_t^2 + W_r^2} = \frac{W_t}{\cos(\phi)} = \frac{1346.68}{\cos(20)}$

$W_n = 1436.3$ lb

$C = 1.38$ in, $V_t = 172.935$ ft/min, $W_t = 1346.68$ lb

$W_r = 490.151$ lb, $W_n = 1436.30$ lb

12. Find the face width (F) and the range

- Nominal face width (Fnom): $F_{nom} = 2P_{xc} = 2(2.34449) = 4.68898$ in

∴ Pick nominal face width (F)

F = 4.6898 in

1. Find Design Power ($P_{des}$)

a) Given Power from the motor → need to find overload factor ($K_o$)

Overload Factor ($K_o$) → Table 9-1 (Problem 7)

TABLE 9-1 Suggested Overload Factors, $K_o$

| Power source | Driven Machine: Uniform | Light shock | Moderate shock | Heavy shock |
|---|---|---|---|---|
| Uniform | 1.00 | 1.25 | 1.50 | 1.75 |
| Light shock | 1.20 | 1.40 | 1.75 | 2.25 |
| Moderate shock | 1.30 | 1.70 | 2.00 | 2.75 |

*Uniform:* Electric motor or constant-speed gas turbine.

*Light shock:* Water turbine, variable-speed drive

*Moderate shock:* Multicylinder engine

Examples of the roughness of driven machines include the following:

*Uniform:* Continuous-duty generator, paper, and film winders.

*Light shock:* Fans and low-speed centrifugal pumps, liquid agitators, variable-duty generators, uniformly loaded conveyors, rotary positive displacement pumps, and metal strip processing.

*Moderate shock:* High-speed centrifugal pumps, reciprocating pumps and compressors, heavy-duty conveyors, machine tool drives, concrete mixers, textile machinery, meat grinders, saws, bucket elevators, freight elevators, escalators, concrete mixers, plastics molding and processing, sewage disposal equipment, winches, and cable reels.

*Heavy shock:* Rock crushers, punch press drives, pulverizers, processing mills, tumbling barrels, wood chippers, vibrating screens, railroad car dumpers, log conveyors, lumber handling equipment, metal shears, hammer mills, commercial washers, heavy-duty hoists and cranes, reciprocating feeders, dredges, rubber processing, compactors, and plastics extruders.

@ uniform power source and moderate shock

$K_o = 1.75$

b) Calculate Design Power ($P_{des}$)

$P_{des} = K_o P = (1.75)(20) = 35$

$P_{des} = 35$ hp

2. Find diametral Pitch ($P_d$) and normal pressure angle ($\phi_n$)

- From Problem 7: $P_d = 5$ (Problem 7)

- Find Normal diametrical pitch ($P_{nd}$)

$P_{nd} = \frac{P_d}{\cos(\psi)} = \frac{5}{\cos(15°)} = 5.17638$

$P_{nd} = 5.17638$ teeth/in

- Find Axial Pitch ($P_x$)

$P_{xc} = \frac{\pi}{P_d \tan(\psi)} = \frac{\pi}{5\tan(15°)} = 2.34492$

$P_x = 2.3449$ in

- Normal Pressure angle

$\phi_n = \arctan[\tan(\phi_t)\cos(\psi)]$

$\phi_n = \arctan[\tan(20)\cos(15)]$

$\phi_n = 19.77° \approx 20°$

∴ Assume $\phi_n = 20°$

3. Choose the Number of teeth of pinion following - From last question $N_p = 18$ (Problem 7)

↳ $N_p = 18$

4. Choose the rotational speed of gear ($N_G$) from a given range (Problem 7)

Given: $180 < n_G < 190$

↳ Pick mid point $n_G = 185$ rpm

5. Find nominal velocity ratio ($VR$)

$VR_{nom} = \frac{n_p}{n_G} = \frac{520}{185} = 2.8108$

$VR_{nom} = 2.8108$

6. Find the number of teeth in Gear ($N_G$) using $VR_{nom}$ and round to nearest integer

$VR_{nom} = \frac{N_G}{N_p} \longrightarrow N_G = (VR_{nom})(N_p) = (2.8108)(18)$

$N_G = 50.59$

$N_G = 51$

7. Find actual velocity ratio ($VR_{actual}$)

$VR_{actual} = \frac{N_G}{N_p} = \frac{51}{18} = 2.8333$

$VR_{actual} = 2.833$

14. Find Elastic Coefficient ($C_p$) → Table 9-7

TABLE 9-7 Elastic Coefficient, $C_p$

Gear material and modulus of elasticity, $E_G$ lb/in² (MPa)

| Pinion material | Modulus of elasticity $E_P$ lb/in² (MPa) | Steel 30 × 10⁶ (207 × 10³) | Malleable iron 25 × 10⁶ (172 × 10³) | Nodular iron 24 × 10⁶ (165 × 10³) | Cast iron 22 × 10⁶ (152 × 10³) | Aluminum bronze 17.5 × 10⁶ (121 × 10³) | Tin bronze 16 × 10⁶ (110 × 10³) |
|---|---|---|---|---|---|---|---|
| Steel | 30 × 10⁶ (207 × 10³) | 2300 (191) | 2180 (181) | 2160 (179) | 2090 (174) | 1960 (162) | 1900 (158) |
| Mall. iron | 25 × 10⁶ (172 × 10³) | 2180 (181) | 2090 (174) | 2070 (172) | 2000 (166) | 1880 (156) | 1820 (151) |
| Nod. iron | 24 × 10⁶ (165 × 10³) | 2160 (179) | 2070 (172) | 2050 (170) | 1990 (165) | 1870 (155) | 1810 (150) |
| Cast iron | 22 × 10⁶ (152 × 10³) | 2090 (174) | 2000 (166) | 1990 (165) | 1900 (158) | 1790 (149) | 1730 (144) |
| Al. bronze | 17.5 × 10⁶ (121 × 10³) | 1960 (162) | 1880 (156) | 1870 (155) | 1790 (149) | 1680 (139) | 1630 (135) |
| Tin bronze | 16 × 10⁶ (110 × 10³) | 1900 (158) | 1820 (151) | 1810 (150) | 1730 (144) | 1630 (135) | 1570 (130) |

Source: Extracted from AGMA Standard 2001-D04, "Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth," with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA 22314.

Note: Poisson's ratio = 0.30, units for $C_p$ are $(lb/in^2)^{0.5}$ or $(MPa)^{0.5}$.

For steel gear and pinion,

$C_p = 2300$

15. Find the AGMA number Ar from the application and pitch line speed

Table 9-5

TABLE 9-5 Recommended AGMA Quality Numbers

| Application | Quality number | Application | Quality number |
|---|---|---|---|
| Cement rotary drum drive | A11 | Small power drill | A9 |
| Cement kiln | A11 | Clothes washing machine | A8 |
| Steel mill drives | A11 | Printing press | A7 |
| Grain harvester | A10 | Computing mechanism | A6 |
| Clarifier | A10 | Automotive transmission | A6 |
| Punch press | A10 | Radar antenna drive | A6 |
| Mining conveyor | A10 | Marine propulsion drive | A5 |
| Paper/pulp making machine | A9 | Aircraft engine drive | A4 |
| Gas motor exhausters | A9 | Gyroscope | A2 |

Machine tool drives and drives for other high-quality mechanical systems

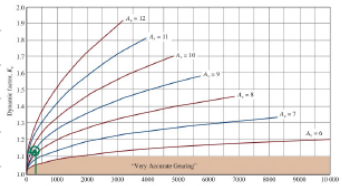
| Pitch line speed (fpm) | Quality number | Pitch line speed (m/s) |
|---|---|---|
| 0–800 | A5 | 0–4 |
| 800–2000 | A6 | 4–11 |
| 2000–4000 | A7 | 11–22 |
| Over 4000 | A8 | Over 22 |

Given AGMA Quality number = A9

∴ Pick A9 for this application

16. Find dynamic factor(Kv) from Figure 9-16 or Table 9-6 based on Vt & Kv

↳ Choose Table method since it is more accurate → $K_v = 1.187729$




(a) U.S. Customary units

FIGURE 9-16 Dynamic factor, $K_v$ (Adapted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

**TABLE 9-6 Analytical Method for Computing Dynamic Factor, $K_v$**

| U.S. units | SI Metric units |
|---|---|
| Given $A_v$ and $v_t$ (ft/min) | Given $A_v$ and $v_t$ (m/s) |
| $B = 0.25(A_v - 5.0)^{0.667}$ | $B = 0.25(A_v - 5.0)^{0.667}$ |
| $C = 50 + 56(1.0 - B)$ | $C = 3.5637 + 3.9914(1.0 - B)$ |
| $K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$ | $K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$ |
| Maximum velocity for a given $A_v$ | Maximum velocity for a given $A_v$ |
| $v_{t\,max} = [C + (14 - A_v)]^2$ | $v_{t\,max} = [C + (14 - A_v)]^2$ |
| $A_v$ — $v_{t\,max}$ (ft/min) | $A_v$ — $v_{t\,max}$ (m/s) |
| 6 — 10 000 | 6 — 50.8 |
| 7 — 8239 | 7 — 41.9 |
| 8 — 6867 | 8 — 34.9 |
| 9 — 5731 | 9 — 29.1 |
| 10 — 4767 | 10 — 24.2 |
| 11 — 3937 | 11 — 20.0 |
| 12 — 3219 | 12 — 16.4 |

→ $K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$

→ $B = 0.25(A_v - 5.0)^{0.667} = 0.25(9-5)^{0.667} = 0.630252$

→ $C = 50 + 56(1 - B) = 50 + 56(1 - 0.62519) = 70.7059$

→ $K_v = \left[\frac{70.7059}{70.7059 + \sqrt{490.095}}\right]^{-0.630252} = 1.118729$

$K_v = 1.118729$

17. Choose Bending Geometry Factor for Gear and Pinion ($J_P$ and $J_G$) from Figure 10-5 ($\phi = 15°$) or 10-6 ($\phi = 20°$)

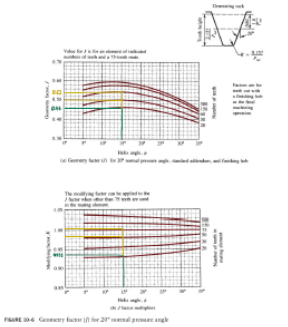



FIGURE 10-6 Geometry factor (J) for 20° normal pressure angle

• For $J_P$ ; GREEN $N_P = 18$

@ $\psi = 15$

$J_P = 0.96$

$J_1 = 0.935$

$J_{PF} = J_P(J_1) = (0.96)(0.935)$

$J_{PF} = 0.430$

• For $J_G$ : YELLOW $N_G = 51$

@ $\psi = 15°$ and $N_G = 51$

$J_G = \frac{J_{G2} - J_{G1}}{N_{G2} - N_{G1}}(N_G - N_{G1}) + J_{G1}$

$J_1 = \frac{J_{2} - J_{1}}{N_{G2} - N_{G1}}(N_G - N_{G1}) + J_{G1}$

$\frac{0.52 - 0.50}{60 - 30}(51 - 30) + 0.50$

$J_1 = \frac{1 + 0.9825}{60 - 30}(51 - 30) + 0.9825$

$J_1 = 0.994125$

$J_G = 0.514$

$J_{G,F} = (J_1)(J_G) = (0.994125)(0.514)$ → $J_{G,F} = 0.511302$

8. Find Pitting Geometry Factor (I)

- Table 10-1 ($\phi = 20°$) and 10-2 ($\phi = 25°$)

**TABLE 10-1 Geometry Factors for Pitting Resistance, $I$, for Helical Gears with 20° Normal Pressure Angle and Standard Addendum**

A. Helix angle $\psi = 15.0°$

| Gear tooth | Pinion tooth 17 | 21 | 26 | 35 | 55 |
|---|---|---|---|---|---|
| 17 | 0.124 | | | | |
| 21 | 0.130 | 0.138 | | | |
| 26 | 0.134 | 0.143 | 0.152 | | |
| 35 | 0.136 | 0.145 | 0.154 | 0.163 | 0.170 |
| 55 | 0.139 | 0.147 | 0.157 | 0.166 | 0.174 |
| 135 | 0.144 | 0.152 | 0.162 | 0.171 | 0.179 |

B. Helix angle $\psi = 25.0°$

| Gear tooth | Pinion tooth 14 | 17 | 21 | 26 | 35 | 55 |
|---|---|---|---|---|---|---|
| 14 | 0.123 | | | | | |
| 17 | 0.137 | 0.146 | | | | |
| 21 | 0.152 | 0.162 | 0.170 | | | |
| 26 | 0.167 | 0.177 | 0.186 | 0.194 | | |
| 35 | 0.187 | 0.198 | 0.208 | 0.216 | 0.224 | |
| 55 | 0.215 | 0.227 | 0.239 | 0.249 | 0.257 | 0.265 |
| 135 | 0.248 | 0.261 | 0.274 | 0.286 | 0.295 | 0.304 |

Source: Extracted from AGMA Standard 908-B89 (9 1989), *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical and Herringbone Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

double interpolate

• At $N_G = 35$

$I = \frac{I_2 - I_1}{N_{P2} - N_{P1}}(N_P - N_{P1}) + I_1 = \frac{0.165 - 0.135}{21 - 13}(18 - 13) + 0.135$

$I = 0.1925$

• At $N_G = 55$

$I = \frac{I_2 - I_1}{N_{P2} - N_{P1}}(N_P - N_{P1}) + I_1 = \frac{0.196 - 0.204}{21 - 13}(18 - 13) + 0.204$

$I = 0.202$

Another interpolation for $N_G = 51$ at $N_P = 18$

$I = \frac{I_2 - I_1}{N_{G2} - N_{G1}}(N_G - N_{G1}) + I_1 = \frac{0.202 - 0.1925}{55 - 35}(51 - 35) + 0.1925$

$I_{final} = 0.1961$

19. Find Pinion Proportion Factor ($C_{pf}$) using Figure 9-12

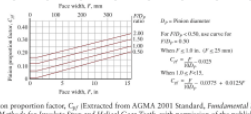



FIGURE 9-12 Pinion proportion factor, $C_{pf}$ (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

$\frac{F}{D_P} = \frac{4.6898}{3.6} = 1.302722 > 0.5$

↗ Use equation for more accuracy

Since $F = 4.6898$ and $1.0 \le F \le 17$

$C_{pf} = \frac{F}{10D_P} - 0.0375 + 0.0125F$

$C_{pf} = \frac{4.6898}{10(3.6)} - 0.0375 + 0.0125(4.6898)$

$C_{pf} = 0.151395$

20. Find Mesh Alignment Factor ($C_{ma}$) using

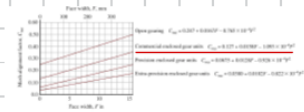



FIGURE 9-13 Mesh alignment factor, $C_{ma}$ (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

→ Use $C_{ma}$ equation for Commercial enclosed gear units

$C_{ma} = 0.127 + 0.0158F - (1.093 \times 10^{-4})F^2$

$C_{ma} = 0.127 + 0.0158(4.6898) - (1.093 \times 10^{-4})(4.6898)^2$

$C_{ma} = 0.198695$

21. Find the Alignment Factor ($K_m$)

$K_m = 1.0 + C_{pf} + C_{ma}$

$K_m = 1.0 + 0.151395 + 0.198695$

$K_m = 1.35$

22. Find Size Factor ($K_S$) → Table 9-2

**TABLE 9-2 Suggested Size Factors, $K_s$**

| Diameter pitch, $P_d$ | Metric module, $m$ | Size factor, $K_s$ |
|---|---|---|
| $\ge 5$ | $\le 5$ | 1.00 |
| 4 | 6 | 1.05 |
| 3 | 8 | 1.15 |
| 2 | 12 | 1.25 |
| 1.25 | 20 | 1.40 |

@ $P_d = 5$ $K_S = 1.00$

$K_S = 1.00$

23. Find Rim Thickness Factor ($K_B$) → Figure 9-14

a) Calculate Backup ratio ($m_B$)

$m_B = \frac{t_R}{h_t}$ $t_R$ = rim thickness; $h_t$ = whole depth of the gear tooth

* If commercially supported gear → can assume that $K_B = 1.0$

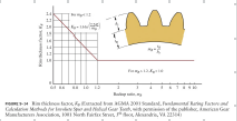



FIGURE 9-14 Rim thickness factor, $K_B$ (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

Using assumption

$K_B = 1.0$

24. Specify Service Factor (SF) → $1.00 \le SF \le 1.50$

↳ typically $F_S = 1.00$ but can change depending on the uncertainty of data

$F_S = 1.00$

25. Specify a Reliability(R) value for Reliability factor ($K_R$)

→ typically pick $R = 0.99$

$R = 0.99$

26. Find Reliability factor ($K_R$) → Table 9-11 using Reliability (R)

**TABLE 9-11 Reliability Factor, $K_R$**

| Reliability | $K_R$ |
|---|---|
| 0.90, one failure in 10 | 0.85 |
| 0.99, one failure in 100 | 1.00 |
| 0.999, one failure in 1000 | 1.25 |
| 0.9999, one failure in 10 000 | 1.50 |

for $R = 0.99$, $K_R = 1.0$

$K_R = 1.0$

27. Specify Design Life Time using → Table 9-12 using design application

→ ALSO GIVEN that lifetime = 20000 hrs (Problem 9-3)

**TABLE 9-12 Recommended Design Life**

| Application | Design life (h) |
|---|---|
| Domestic appliances | 1000–2000 |
| Aircraft engines | 1000–4000 |
| Automotive | 1500–5000 |
| Agricultural equipment | 3000–6000 |
| Elevators, industrial fans, multipurpose gearing | 8000–15 000 |
| Electric motors, industrial blowers, general industrial machines | 20 000–30 000 |
| Pumps and compressors | 40 000–60 000 |
| Critical equipment in continuous 24-h operation | 100 000–250 000 |

Source: Eugene A. Avallone and Theodore Baumeister III, eds. Marks' *Standard Handbook for Mechanical Engineers*, 9th ed. New York: McGraw-Hill, 1986.

For electric motors, design life time

20000 – 20000 hr

Choose design life time = 20,000 hr

28. Find the number of loading cycles ($N_{CP}$ and $N_{CG}$)

$N_{CP} = 60(L_{lifetime})(n_p)n_p = 60(20000)(520)(1) = 6.24 \times 10^8$ cycle

$N_{CG} = 60(L_{lifetime})(n_p)n_{G,a} = 60(20000)(183.532)(1) = 2.2023 \times 10^8$ cycle

* Note: use actual speed output ($n_{G,a}$) & q = # of load applications per revolution (q=1)

$N_{CP} = 6.24 \times 10^8$ cycles, $N_{CG} = 2.20 \times 10^8$ cycles

29. Find Bending stress cycle factor for Pinion and Gear ($Y_{NP}$, $Y_{NG}$) → Figure 9-21

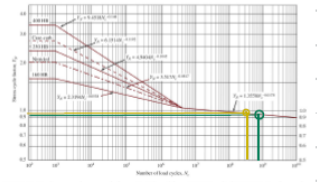


FIGURE 9-21 Bending strength stress cycle factor, $Y_N$ (Adapted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

- For Pinion ($Y_{CP}$) GREEN

@ $N_{CP} = 6.24 \times 10^8$ cycles

$Y_{NP} = 1.3558\,N_C^{-0.0178}$

$Y_{NP} = 1.3558\,(6.24 \times 10^8)^{-0.0178}$

$Y_{NP} = 0.945476$

- For Gear ($Y_{CG}$) YELLOW

@ $N_G = 2.20 \times 10^8$ cycles

$Y_{NG} = 1.3558\,N_C^{-0.0178}$

$Y_{NG} = 1.3558\,(2.20 \times 10^8)^{-0.0178}$

$Y_{NG} = 0.963165$

30. Find Pitting resistance stress cycle factor ($Z_{NP}$, $Z_{NG}$) → Figure 9-22

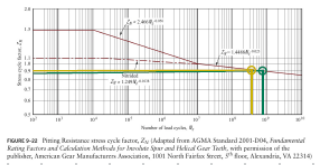


FIGURE 9-22 Pitting Resistance stress cycle factor, $Z_N$ (Adapted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

- For Pinion ($Y_{CP}$) GREEN

@ $N_{CP} = 6.24 \times 10^8$ cycles

$Z_{NP} = 1.4488\,N_C^{-0.023}$

$Z_{NP} = 1.4488\,(6.24 \times 10^8)^{-0.023}$

$Z_{NP} = 0.909325$

- For Gear ($Y_{CG}$) YELLOW

@ $N_G = 2.20 \times 10^8$ cycle

$Z_{NG} = 1.4488\,N_C^{-0.023}$

$Z_{NG} = 1.4488\,(2.20 \times 10^8)^{-0.023}$

$Z_{NG} = 0.931392$

31. Calculate expected bending stresses in the pinion ($S_{tP}$) and gear ($S_{tG}$)

$S_{tP} = \dfrac{W_t P_d}{F J_P} K_O K_S K_m K_B K_V$

Recall: $W_t = 1346.68$ lb, $F = 4.6898$, $J_P = 0.430$, $P_d = 5$, $K_O = 1.75$, $K_S = 1.00$, $K_m = 1.35$, $K_B = 1.0$, $K_V = 1.118729$

$S_{tP} = \dfrac{(1346.68)(5)}{(4.6898)(0.473)}(1.75)(1.00)(1.35)(1.00)(1.118729)$

$S_{tP} = 8376.81$ psi

$S_{tG} = \dfrac{J_P}{J_G} S_{tP}$

Recall: $J_P = 0.430$, $J_G = 0.511702$

$S_{tG} = \dfrac{0.430}{0.511702}(8066.56)$

$S_{tG} = 6783.9$ psi

32. Find an "adjusted" expected bending stress in gear and Pinion ($S_{at}G$ and $S_{at}P$)

Recall: $K_R = 1.0$, $S_F = 1.0$, $Y_{NP} = 0.909325$, $Y_{NG} = 0.945476$

$S_{at}P > \dfrac{K_R(S_F)}{Y_{NP}} S_{tP}$ → $S_{at}P > \dfrac{(1.0)(1.0)}{0.909325}(8376.81) = 9212.12$ psi → $S_{at}P > 9212.12$ psi

$S_{at}G > \dfrac{K_R(S_F)}{Y_{NG}} S_{tG}$ → $S_{at}G > \dfrac{(1.0)(1.0)}{0.945476}(6783.9) = 7175.27$ psi → $S_{at}G > 7175.29$ psi

33. Calculate expected Pitting Contact Stress ($S_C$)

Recall $C_p = 2300$, $I = 0.1961$ (From step #14)

$S_C = C_p\sqrt{\dfrac{W_t K_O K_S K_m K_V}{F D_P (I)}}$ → $S_C = (2300)\sqrt{\dfrac{(1346.68)(1.75)(1.0)(1.35)(1.118729)}{(4.6898)(3.6)(0.1961)}}$

$S_C = 75414.5$ psi

Note

Pitting Contact Stress of Gear and Pinion are equal so

$S_{CP} = S_{CG} = S_C = 75414.5$ psi

34. Find the "adjusted" value of $S_C$ for gear and pinion ($S_{ac}G$, $S_{ac}P$)

$S_{ac}P > \dfrac{K_R(S_F)}{Z_{NP}} S_{CP}$ → $S_{ac}P > \dfrac{(1.0)(1.0)}{0.909325}(75414.5) = 82939.6$ → $S_{ac}P > 82934.6$ psi

$S_{ac}G > \dfrac{K_R(S_F)}{Z_{NG}} S_{CG}$ → $S_{ac}G > \dfrac{(1.0)(1.0)}{0.931392}(75414.5) = 80969.7$ → $S_{ac}G > 80969.7$ psi

25. Calculate the required HB (Brinell hardness number) using Figure 19-18 (bending stress) and Figure 19-19 (Pitting contact stress)

Note: For most of the time → ONLY need to check HB of Pitting Contact stress as stress is larger

→ Figure 9-19 for steel Grade 1 & 2

- After finding HB of stress in grade 1 & 2 → pick the largest HB

Note from Figure 9-19
- required 180 < HB < 400
- desirable HB range 250 < HB < 400
→ if HB < 180 → make gear more compact
→ if HB > 400 → no value can be considered

Figure 19-18

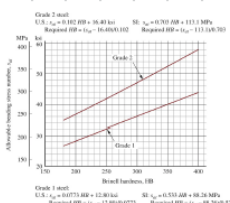



FIGURE 9-18 Allowable bending stress number for through-hardened steel gears, $\sigma_{at}$ (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

Figure 19-19

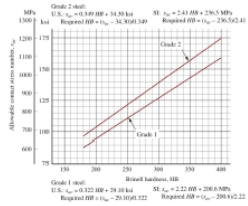



FIGURE 9-19 Allowable contact stress number for through-hardened steel gears, $\sigma_{ac}$ (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

Grade 1: $HB = \frac{S_{ac}}{1000} - 29.10 \,/\, 0.322$ → Gear $HB_G = \frac{\frac{80969.7}{1000} - 29.10}{0.321}$

$HB_{G,\,grade\,1} = 161.086$

→ Pinion $HB_P = \frac{\frac{82974.6}{1000} - 29.10}{0.322}$

$HB_{P,\,grade\,1} = 167.188$

Grade 2: $HB = \frac{S_{ac}}{1000} - 34.30 \,/\, 0.349$ → Gear $HB_G = \frac{\frac{80969.7}{1000} - 34.30}{0.749}$

$HB_{G,\,grade\,2} = 132.005$

→ Pinion $HB_P = \frac{\frac{82974.6}{1000} - 34.30}{0.349}$

$HB_{P,\,grade\,2} = 139.354$

Among 4 HB, pick $HB_{P,\,grade\,1}$ since Pinion have more critical value of $S_{ac}$ (Pitting Contact stress)

$HB_{P,\,grade\,1} = 167.188$

26. Use Appendix 3, 4, 5 to find material that satisfy the required HB for gear and pinion

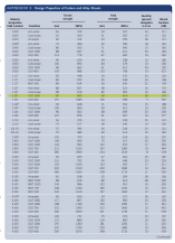
Note: Should use same material for Gear & Pinion

HB > 167.188

Gear and Pinion have same material at $\sigma_{tensile}$ = 87 ksi, $\sigma_{yield}$ = 60 ksi, 28% elongation

① HB 174: SAE 1137 OQT 1300

Appendix 3



- Specify Heat Treatment → Appendix 4 (since material chosen (Steel SAE 1137 heat treatment is not listed in Appendix 4, heat treatment can be found online)

reference: https://steelheattreatment.com/d/1137-carbon-steel/

37. Calculate safety factors for gears for both bending stress and Pitting contact stress and check if $1.0 \leq F \leq 1.5$

↳ with the chosen material ⟶ ① HB 174 ; SAE 1137 OQT 1300 at $\sigma_{tensile}$ = 87 ksi, $\sigma_{yield}$ = 80 ksi ; 28 % elongation

↳ use figure 9-18 (bending) and 9-19 (Pitting Contact) to get $S_{at}$ and $S_{ac}$

Figure 19-18

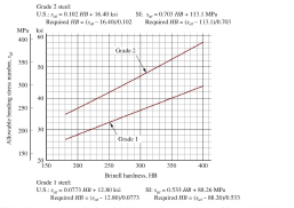



FIGURE 9-18 Allowable bending stress number for through-hardened steel gears, $s_{at}$ (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

Figure 19-19

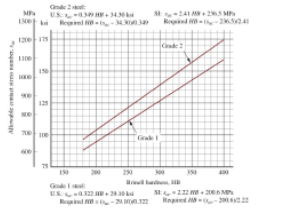



FIGURE 9-19 Allowable contact stress number for through-hardened steel gears, $s_{ac}$ (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314) [Reference 6]

$S_{at} = 0.102\,HB + 16.4$ ksi

$= 0.102(174) + 16.4$

$S_{at} = 34.148$ ksi

$S_{ac} = 0.349\,HB + 34.3$ ksi

$= 0.349(174) + 34.3$

$S_{ac} = 95.026$ ksi

Final Safety Factor

Final Safety Factor

Recall: Bending

$S_{at}P = 9217.12$ psi  $K_R = 1.0$

$S_{at}G = 3175.27$

$Y_{NP} = 0.909325$

$Y_{NG} = 0.963165$

$S_{tP} = 8376.81$

$S_{tG} = 6783.9$

Pitting Contact

$S_{ac}P = 87924.6$ psi

$S_{ac}G = 80969.7$ psi

$Z_{NP} = 0.909325$

$Z_{NG} = 0.931392$

$S_{cG} = 75414.5$

$S_{cP} = 75414.5$

• For Bending

$$SF_P = \frac{(S_{at}P)(Y_{NP})}{(S_{tP})(K_R)} = \frac{(34.148\times10^3)(0.909325)}{(8376.81)(1.0)} = 3.70686$$

$$SF_G = \frac{(S_{at}G)(Y_{NG})}{(S_{tG})(K_R)} = \frac{(31175\times10^3)(0.963165)}{(6783.9)(1.0)} = 4.84827$$

$SF_P$ & $SF_G \geq 1.00$ ⟶ work

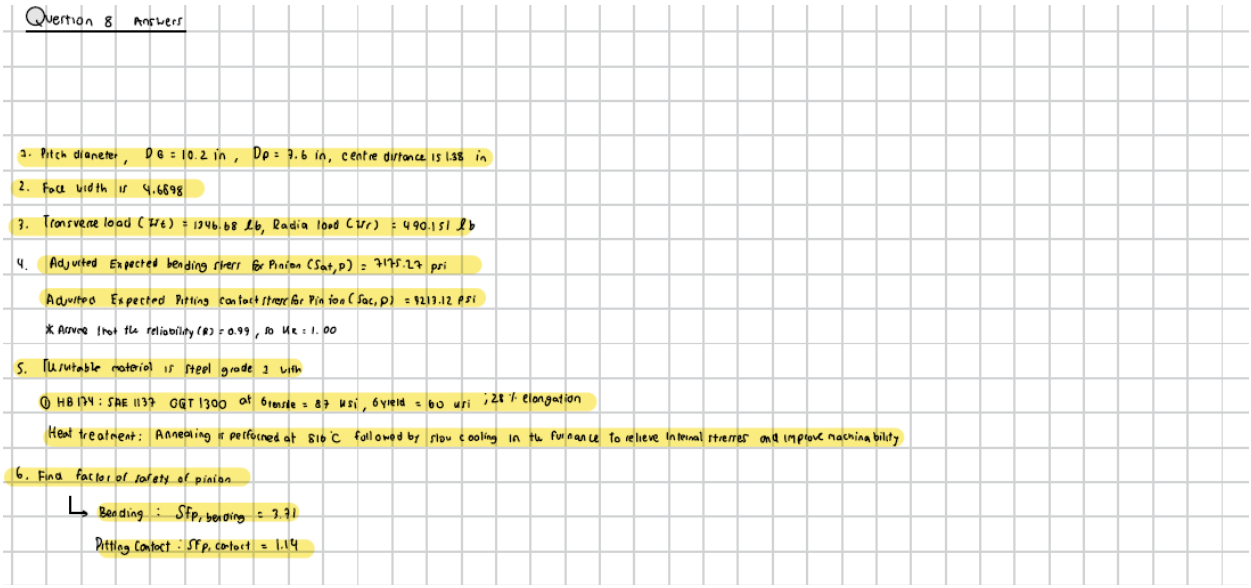
• For Pitting contact

$$SF_P = \frac{S_{ac}P\,Z_{NP}}{S_{cP}K_R} = \frac{(95.026\times10^3)(0.909325)}{(75414.5)(1.0)} = 1.14579$$

$$SF_G = \frac{S_{ac}G\,Z_{NG}}{S_{cG}K_R} = \frac{(95.026\times10^3)(0.931392)}{(75414.5)(1.0)} = 1.1736$$

$SF_P$ : $SF_G \geq 1.00$ ⟶ work

* Factor of safety for Bending stress exceed the limit of 1.5, however since the value of Pitting contact stress is more critical, need to make sure that Factor of safety of Pitting contact stress is within 1.00 and 1.50 range. In this question, FS of both gear and pinion are within 1.00 & 1.50




**Problem 9 (Mott 8-52 and 10-18 modified)**

A wormgear set has a single-thread worm with a pitch diameter of 1.250 inches, a diametrical pitch of 10, and a normal pressure angle of 14.5°. The worm meshes with a wormgear having 40 teeth and a face width of 0.625 inches. The wormgear set is transmitting 924 lb-in. of torque at the output shaft which is rotating at 30 rpm. Compute:

a) The gear ratio and the input speed
b) the lead and lead angle
c) gear pitch diameter,
d) the center distance,
e) the sliding speed
f) the forces on the gear (tangential, axial, radial)
g) the friction force and friction power
h) the required input power
i) the wormgear efficiency
j) the stress on the gear teeth

If the worm is hardened steel and the worm is chilled bronze, determine
k) the rated tangential load
l) and whether the design is satisfactory to resisit pitting.

52. **Worm Gearing – Given data:** $D_w = 1.250$ in; $N_w = 1$; $P_d = 10$; $\phi_n = 14.5°$



$N_G = 40$; $F = 0.625$ in Single Thread

**Lead** = Axial Pitch = Circular Pitch = $\pi/P_d = \pi/10 = \mathbf{0.3142\ in}$

**Lead Angle** = $\boldsymbol{\lambda} = tan^{-1}\left(\frac{L}{\pi D_w}\right) = tan^{-1}\left(\frac{0.3142}{\pi(1.250)}\right) = \mathbf{4.57°}$

**Addendum** = $\boldsymbol{a} = \frac{1}{P_d} = \frac{1}{10} = \mathbf{0.100\ in}$; **Dedendum** = $\boldsymbol{b} = \frac{1.157}{P_d} = 0.1157$ in

**Worm Outside Diameter** = $\boldsymbol{D_{ow}} = D_w + 2a = 1.250 + 2(0.100) = \mathbf{1.450\ in}$

**Worm Root Diameter** = $\boldsymbol{D_{rw}} = D_w - 2b = 1.250 - 2(0.1157) = \mathbf{1.0186\ in}$

**Gear Pitch Diameter** = $\boldsymbol{D_G} = N_G/P_d = 40/10 = \mathbf{4.000\ in}$

**Center Distance** = $\boldsymbol{C} = (D_G + D_w)/2 = (4.000 + 1.250)/2 = \mathbf{2.625\ in}$

**Velocity Ratio** = $\boldsymbol{VR} = \boldsymbol{N_G}/N_w = 40/1 = \mathbf{40}$

18. [Shown here are data from the solution to Problem 8-52 that has the same requirements] $T_o = 924$ lb · in, $n_G = 30$ rpm

Forces: $W_{tG} = W_{xw} = \frac{T_o}{D_G/2} = \frac{924\ lb{\cdot}in}{2.00\ in} = 462$ lb

Pitch Line Speed of Gear $v_{tG} = \pi D_G n_G / r = \pi(4.00)(30)/12 = 31.4$ ft/min

Sliding Velocity $= v_s = v_{tG} / \sin\lambda = 31.4 / \sin(4.57) = 394$ ft/min

From Figure 10-25; $\mu = 0.0323$ (Computed from Equation $10-27$)

$W_{xG} = W_{tW} = 462 \times \frac{\cos(14.5)\sin(4.57)+0.0323\cos(4.57)}{\cos(14.5)\cos(4.57)-0.0323\times\sin(4.59)} = 53$ lb (Equation 10-30)

$W_{rG} = W_{rW} = \frac{462\times\sin(14.5)}{\cos(4.57)\cos(14.5)-0.0323\sin(4.57)} = 120$ lb (Equation 10-31)

Friction Force $= W_f = \frac{(0.0323)(462)}{\cos(4.57)\cos(14.5)-0.0323\sin(4.57)} = 15.5$ lb (Equation 10-32)

Friction Power Loss $= P_L = \frac{v_s W_f}{33\,000} = \frac{(394)(15.5)}{33\,000} = 0.185$ hp

Input Power $= P_{in} = P_o + P_L = \frac{T_o n_G}{63\,000} + 0.185 = 0.44 + 0.185 = 0.625$ hp

Efficiency $= P_o / P_{in} \times 100\% = 70.4\%$

Input Speed $= n_G \times v_R = (30)(40) = 1200$ rpm

Stress $\sigma_G = \frac{W_d}{yFp_n} = \frac{W_{tG}P_d}{K_v yF\pi\cos\lambda} = \frac{(462)(10)}{(0.974)(0.100)(0.625)(\pi)(\cos 4.57)} = 24\,235$ psi

Pitting $W_{tR} = C_s D_G^{0.8} F_e C_m C_v = (1000)(4.00)^{0.8}(0.625)(0.814)(0.427) = 659$ lb

Because $W_{tR} > W_{tG}$ – Ok for pitting

1) single thread → 40 teeth

$$VR = \frac{N_G}{N_W} = \frac{40}{1}$$

**40 : 1 reduction**

$$n_{in} = n_{out} \cdot VR = 30 \cdot 40$$

**$n_{in}$ = 1200 rpm**

2) Lead, $L = N_W \cdot P_x$ Eq. 8-23

↑ axial pitch of worm = circular pitch of gear, $p$

$$p = \pi \frac{D_G}{N_G}$$ Eq. 8-20

circular pitch, $p = \pi \cdot \frac{1}{P_d}$ Eq. 8-22 → $p = P_x = \frac{\pi}{10\ T/in} = 0.31416\ in/T$

**$L$ = 0.31416 in** for $N_W = 1$

Lead angle: $\tan\lambda = \frac{L}{\pi D_w}$ Eq. 8-24 ($\pi D_w$ = circumference)

$$\lambda = \tan^{-1}\left(\frac{0.31416\ in}{\pi \cdot 1.25\ in}\right)$$

**$\lambda = 4.574°$**

3) $D_G = \frac{N_G}{P_d} = \frac{40\ T}{10\ T/in} = 4\ in$

**$D_G$ = 4.0 in**

4) $CD = \frac{D_G}{2} + \frac{D_w}{2} = \frac{4.0}{2} + \frac{1.25}{2}$

**CD = 2.625 in**

5) Sliding Speed: $V_S = \frac{V_w}{\cos\lambda}$ [Eq. 10-25]

$$V_w = \frac{\pi d_w n_w}{12} = \frac{\pi}{12}(1.25\ in)(1200\ rpm) = 392.7\ ft/min$$

$$V_S = \frac{392.7}{\cos(4.574°)}$$

**$V_S$ = 393.95 fpm**

6) $W_{t_G} = 2T_o / D_G = 924\ \text{lb-in} / (4\ \text{in}/2) = 462\ \text{lb}$ $\boxed{W_{t_G} = 462\ \text{lb}}$

[Eq. 10-29]

$$W_{a_G} = W_{t_G} \frac{\cos\phi_n \sin\lambda + \mu\cos\lambda}{[\cos\phi_n \cos\lambda - \mu \sin\lambda]} = \frac{462\,(\cos 14.5^\circ \sin 4.574^\circ + 0.0324 \cos 4.574^\circ)\ \text{lb}}{(\cos(14.5^\circ)\cos 4.574^\circ - 0.0324 \sin 4.574^\circ)}$$

[Eq. 10-30]

$\boxed{W_{a_G} = 52.6\ \text{lb}}$

$$W_{r_G} = \frac{W_{t_G} \sin\phi_n}{\cos\phi_n \cos\lambda - \mu \sin\lambda} = \frac{462 \sin(14.5^\circ)\ \text{lb}}{\cos 14.5^\circ \cos 4.574^\circ - 0.0324 \sin 4.574^\circ} = 120.185$$

[Eq. 10-31]

$\boxed{W_{r_G} = 120.2\ \text{lb}}$

$\mu = 0.103\, e^{(-0.110\, V_s^{0.450})} + 0.012$ [Eq. 10-27] $\mu = 0.0324$

7) $W_f = \dfrac{\mu W_{t_G}}{\cos\lambda \cos\phi_n - \mu \sin\lambda} = \dfrac{0.0324\ (462\ \text{lb})}{\cos 4.574 \cos 14.5 - 0.0324 \sin 4.574}$ $\boxed{W_f = 15.55\ \text{lb}}$

[Eq. 10-32]

$P_L = \dfrac{V_s W_f}{33\,000} = \dfrac{393.95\ \text{fpm} \cdot (15.55\ \text{lb})}{33\,000}$ $\boxed{P_L = 0.1857\ \text{hp}}$ lost to friction

(friction power) [Eq. 10-33]

8) INPUT POWER

$T_o = \dfrac{63\,025\,(P_o)}{n_G}$ [Eq. 10-28]

$P_i = P_o + P_L$

[Eq. 10-34]

$P_o = \dfrac{T_o \cdot n_G}{63025} = \dfrac{924\ \text{lb-in} \cdot 30\ \text{rpm}}{63025}$ $P_o = 0.4398\ \text{hp}$

$P_i = (0.44 + 0.1857)\ \text{hp}$

$\boxed{P_i = 0.626\ \text{hp}}$

9) EFFICIENCY

$\eta = \dfrac{P_o}{P_i} = \dfrac{0.44}{0.626}$ $\boxed{\eta = 0.703 = 70.3\ \%}$

[Eq. 10-35] OR [Eq. 10-36]

10) STRESS ON GEAR TEETH:

$\sigma = \dfrac{W_d}{y F p_n}$ [Eq. 10-37]

$V_{t_G} = \dfrac{\pi D_G n_G}{12}$ [10.41]

$V_{t_G} = 30\ \text{rpm} \cdot \dfrac{2\pi\ \text{rad}}{\text{rev}} \cdot \dfrac{4.0\ \text{in}}{2} \cdot \dfrac{1\ \text{ft}}{12\ \text{in}} = 31.42\ \text{ft/min}$

$K_v = 1200 / (1200 + V_{t_G}) = 0.9745$ [10-40]

$W_d = W_{t_G} / K_v$ [10-39]

$= 462\ \text{lb} / 0.9745$ $W_d = 474\ \text{lb}$

$y = 0.100$ [Table 10-5]

$F = 0.625$ in

(OR $p_n = \frac{\pi \cdot D}{40}$ circumference / # of teeth)

$P_n = P \cos\lambda = 0.3142 \cos(4.574°) = 0.31316$ in

$\sigma = \frac{474 \text{ lb}}{(0.100)(0.625 \text{ in})(0.31316)}$ $\quad \sigma = 24\,217.7$ psi

$\boxed{\sigma = 24.22 \text{ ksi}}$

WORM = hardened steel
GEAR = chilled bronze

1) RATED TANGENTIAL LOAD : $W_{t\,all} = C_s D_m^{0.8} F_e C_m C_v$ [Eq. 10-42]

$C_s = 1000$ (see 10-44, pg. 462)

$D_G = 4.0$ in

$F_e = 0.625$ in

$C_m = 0.815$ @ $m_G = 40$ [Fig. 10-28]

$C_v = 0.43$ @ $V_s = 394$ fpm [Fig. 10-29]

$W_{t\,all} = 1000(4)^{0.8}\,0.625\,(0.815)(0.43)$

$\boxed{W_{t\,all} = 664 \text{ lb}}$

OK since $> W_{tG}$

2) Design is satisfactory and should resist pitting.

because $W_{t\,all} > W_{tG}$

$664 \text{ lb} > 462 \text{ lb}$ OK

**Problem 10 (Mott 10-14 modified)**

A straight bevel gear pair has the following data: $N_P = 15$, $N_G = 45$, $P_d = 10$ teeth/inch, 20° pressure angle, face width, F = 1.25 inch. The pinion is driven at 300rpm by a single cylinder gas engine delivering 3HP. The load is a concrete mixer producing moderate shock. The recommended life for the machine is 1000 hours of operation. Assume the gears are cantilevered (not straddle mounted) and of commercial gear quality ($A_v = 8$, $Q_v = 9$).

Design a set of bevel gears specifying:

a) The pitch diameters of each gear and the associated cone angles
b) The transverse, radial and axial forces
c) Determine the bending and contact stress numbers for the pinion only. Be sure to list any assumptions in this analysis.

d) Determine a suitable material and hardness for the pinion (you can assume the same material will be selected for the gear). Specify the alloy and heat treatment.
e) What is the final safety factor for the design (pinion only)?

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Design a set of bevel gears specifying:

1. The pitch diameters of each gear and the associated cone angles
2. The transverse, radial and axial forces
3. Determine the bending and contact stress numbers for the pinion only. Be sure to list any assumptions in this analysis.
4. Determine a suitable material and hardness for the pinion (you can assume the same material will be selected for the gear). Specify the alloy and heat treatment.
5. What is the final safety factor for the design (pinion only)?

| GIVEN: | | F = 1.25 in |
|---|---|---|
| Np = 15 | | hp = 300 rpm |
| NG = 45 | | P = 3 hp |
| Pd = 10 teeth/in | | Lifetime = 1000 hr |
| $\phi$ = 20° | | Av = 8, Qv = 9 |

1. Find Design Power (Pdes)

a) Given Power from the motor ⟶ need to find overload factor (Ko)

Overload Factor (Ko) ⟶ Table 9-1 (Problem 9)

TABLE 9–1 Suggested Overload Factors, $K_o$

| | Driven Machine | | | |
|---|---|---|---|---|
| Power source | Uniform | Light shock | Moderate shock | Heavy shock |
| Uniform | 1.00 | 1.25 | 1.50 | 1.75 |
| Light shock | 1.20 | 1.40 | 1.75 | 2.25 |
| Moderate shock | 1.30 | 1.70 | 2.00 | 2.75 |

Uniform: Electric motor or constant-speed gas turbine.
Light shock: Water turbine, variable-speed drive.
Moderate shock: Multicylinder engine.

Examples of the roughness of driven machine include the following:
Uniform: Continuous-duty generators, paper, and film winders.
Light shock: Fans and low-speed centrifugal pumps, liquid agitators, variable-duty generators, uniformly loaded conveyors, rotary positive displacement pumps, and machine tool drives.
Moderate shock: High-speed centrifugal pumps, reciprocating pumps and compressors, heavy-duty conveyors, machine tool drives, cement mixers, textile machinery, mine grinders, saws, hoist devices, freight elevators, escalators, concrete mixers, plastics molding and processing, rotary kilns, and uniformly loaded elevators.
Heavy shock: Rock crushers, punch press drives, pulverizers, processing mills, molding barrels, wood chippers, vibrating screens, railroad car dumpers, log conveyors, lumber handling equipment, hoists, cranes, rubber handling equipment, heavy-duty loaders, oil-field pumping equipment, log barkers, dredges, rubber processing, compressors, and plastics extruders.

@ Moderate shock for power source & driven machine

$K_o = 1.50$

b) Calculate Design Power (Pdes)

$P_{des} = K_o P = (1.50)(3) = 4.50$

$P_{des} = 4.5$ hp

2. Find velocity ratio (VR)

$VR_{actual} = \frac{N_G}{N_P} = \frac{45}{15} = 3$

$VR_{actual} = 3$

4. Find rotational speed of gear ($n_G$)

$VR_{actual} = \frac{N_G}{N_P} = \frac{n_P}{n_G} \longrightarrow n_G = \frac{n_P}{VR_{actual}}$

$n_G = \frac{300}{3} = 100$ rpm

$n_G = 100$ rpm

6. Find Pitch diameter ($D_P$ & $D_G$)

$D_G = \frac{N_G}{P_d} = \frac{45}{10} = 4.5$ in

$D_P = \frac{N_P}{P_d} = \frac{15}{10} = 1.5$ in

$D_G = 4.5$ in, $D_P = 1.5$ in

7. Compute the pitch cone angle for gear ($\Gamma$) and Pinion ($\gamma$)

$\tan(\gamma) = \frac{N_P}{N_G} \Rightarrow \gamma = \arctan\left[\frac{N_P}{N_G}\right] = \arctan\left[\frac{15}{45}\right] = 18.43°$

$\tan(\Gamma) = \frac{N_G}{N_P} \Rightarrow \Gamma = \arctan\left[\frac{N_G}{N_P}\right] = \arctan\left[\frac{45}{15}\right] = 71.57°$

$\therefore \Gamma = 71.57°$ and $\gamma = 18.43°$

8. Compute outer cone distance for gear and pinion ($A_{o,G}$, $A_{o,P}$)

$A_{o,G} = \frac{D_G}{2\sin(\Gamma)} = \frac{4.5}{2\sin(71.57)} = 2.37164$ in

$A_{o,P} = \frac{D_P}{2\sin(\gamma)} = \frac{1.5}{2\sin(18.43)} = 2.37232$ in

$A_{o,G} = 2.37164$ in, $A_{o,P} = 2.37232$ in

9. Find Pitch Line velocity ($V_t$)

$V_t = \frac{\pi D_G n_G}{12} = \frac{\pi D_P n_P}{12} = \frac{\pi(4.5)(100)}{12} = 117.81$ ft/min

10. Find the torque on the pinion & gear ($T_P$ & $T_G$)

$T_P = 63000\left(\frac{P}{n_P}\right) = 63000\left(\frac{3}{300}\right) = 630$ lbf·in

$T_G = 63000\left(\frac{P}{n_G}\right) = 63000\left(\frac{3}{100}\right) = 1890$ lbf·in

$T_P = 630$ lbf·in, $T_G = 1890$ lbf·in

11. Find the mean radius of gear ($R_m$) and Pinion ($r_m$)

$R_{m,G} = \frac{D_G}{2} - \frac{F}{2}\sin(\Gamma) = \frac{4.5}{2} - \frac{1.25}{2}\sin(71.57) = 1.65726$ in

$r_{m,P} = \frac{D_P}{2} - \frac{F}{2}\sin(\gamma) = \frac{1.5}{2} - \frac{1.25}{2}\sin(18.43) = 0.552409$ in

$R_{m,G} = 1.165726$ in, $r_{m,P} = 0.552409$ in

12. Find different load on Gear and Pinion

- Transmitted load ($W_t$)

$W_t = \frac{126000P}{nD} = \frac{33000P}{V_t} = \frac{T_P}{r_{m,P}} = \frac{T_G}{R_{m,G}} = \frac{12600(3)}{(300)(1.5)} = 840$ lbf

- Radial load ($W_{r,G}$, $W_{r,P}$)

$W_{r,G} = W_t \tan(\phi)\cos(\Gamma) = (840)\tan(20)\cos(71.57) = 96.6568$ lbf

$W_{r,P} = W_t \tan(\phi)\cos(\gamma) = (840)\tan(20)\cos(18.43) = 290.054$ lbf

- Axial load ($W_{x,G}$, $W_{x,P}$)

$W_{x,G} = W_t \tan(\phi)\sin(\Gamma) = (840)\tan(20)\sin(71.57) = 290.054$ lbf

$W_{x,P} = W_t \tan(\phi)\sin(\gamma) = (840)\tan(20)\sin(18.43) = 96.6568$ lbf

$W_t = 840$ lbf → $W_{r,G} = 96.6568$ lbf and $W_{x,G} = 290.054$ lbf
$W_{r,P} = 290.054$ lbf and $W_{x,P} = 96.6568$ lbf

13. Find Size factor - Table 9-2

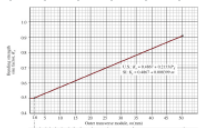


FIGURE 10-21 Size factor for bending stress, $K_s$, for bevel gears (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

- if $P_d \geq 0.5 \longrightarrow K_s = 0.50$
- if $P_d < 0.5 \longrightarrow$ use equation from Table 9-2

$K_s = 0.4867 + \frac{0.2132}{P_d}$

$= 0.4867 + \frac{0.2132}{16}$

$K_s = 0.500031 \approx 0.5$

14. Find load Distribution factor ($K_m$) → Figure 10-19 → Can also use equation to the right

a) Find $K_{mb}$ given that

$K_{mb} = 1.00$ for both gears straddle mounted
$K_{mb} = 1.10$ for one gear straddle mounted
$K_{mb} = 1.25$ for neither gear straddle mounted

(Gear not straddle mount)

* normally, can assure that both gears are

b) Find $K_m$ using

$K_m = K_{mb} + 0.0036F^2$

$K_m = 1.25 + 0.0036(1.25)^2$

$K_m = 1.25563$

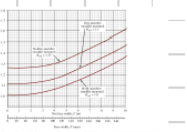


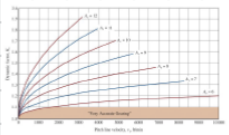
FIGURE 10-19 Load-distribution factors, $K_m$, for commercial enclosed units (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

15. Find the AGMA Quality Number - Table 9-5 OR Given

Given: $A_v = 8$

16. Find Dynamic Factor (Crowned spur gear)

- either use Figure 9-16 or Table 9-6



$$K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$$

$$B = 0.25(A_v - 5.0)^{0.667} = 0.25(8-5)^{0.667} = 0.520211$$

$$C = 50 + 56(1 - B) = 50 + 56(1 - 0.520211) = 76.8682$$

$$K_v = \left[\frac{76.8682}{76.8682 + \sqrt{117.81}}\right]^{-0.520211}$$

$K_v = 1.07113$

TABLE 9-6 Analytical Method for Computing Dynamic Factor, $K_v$

| U.S. units | | SI Metric units | |
|---|---|---|---|
| Given $A_v$ and $v_t$ (ft/min) | | Given $A_v$ and $v_t$ (m/s) | |
| $B = 0.25(A_v - 5.0)^{0.667}$ | | $B = 0.25(A_v - 5.0)^{0.667}$ | |
| $C = 50 + 56(1.0 - B)$ | | $C = 3.5637 + 3.9914(1.0 - B)$ | |
| $K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$ | | $K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$ | |
| Maximum velocity for a given $A_v$ | | Maximum velocity for a given $A_v$ | |
| $v_{t,max} = [C + (14 - A_v)]^2$ | | $v_{t,max} = [C + (14 - A_v)]^2$ | |
| $A_v$ | $v_{t,max}$ (ft/min) | $A_v$ | $v_{t,max}$ (m/s) |
| 6 | 10 000 | 6 | 50.8 |
| 7 | 8239 | 7 | 41.9 |
| 8 | 6867 | 8 | 34.9 |
| 9 | 5731 | 9 | 29.1 |
| 10 | 4767 | 10 | 24.2 |
| 11 | 3937 | 11 | 20.0 |
| 12 | 3219 | 12 | 16.4 |

17. Find Geometry factor for Bending strength → Figure 10-15

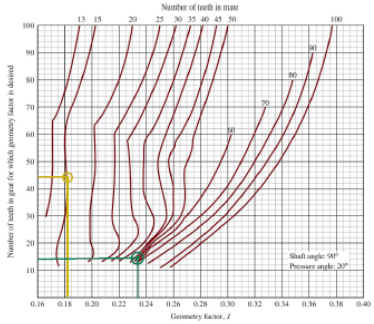



FIGURE 10-15 Geometry factor, J, for straight bevel gears with 20° pressure angle and 90° shaft angle (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

- Gear: $N_G$ = 45 (y axis), $N_P$ = 15 (top x axis) YELLOW

$J_G = 0.282$

- Pinion: $N_P$ = 15 (y axis), $N_G$ = 45 (top x axis) GREEN

$J_P = 0.274$

End of Bending Stress number content

Start the content for Allowable Stress number

18. Specify service factor (SF) → $1.00 \le SF \le 3.50$

usually pick SF = 1.0 $SF = 1.0$

19. Determine Hoverreliability (R), reliability factor ($K_R$), Contact reliability factor ($C_R$)

a) Reliability (R) → usually pick 0.99 $R = 0.99$

b) Reliability Factor ($K_R$) → Table 10-3

TABLE 10-3 Reliability Factors for Allowable Bending and Contact Stresses

| Reliability # | Interpretation | Reliability factors: Bending $K_R$ | Reliability factors: Contact $C_R$ |
|---|---|---|---|
| 0.9 | Fewer than one failure in 10 | 0.85 | 0.92 |
| 0.99 | Fewer than one failure in 100 | 1.00 | 1.00 |
| 0.999 | Fewer than one failure in 1000 | 1.25 | 1.12 |
| 0.9999 | Fewer than one failure in 10 000 | 1.50 | 1.22 |

Source: Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.

For a reliability of 0.99 (or 99%.)

$K_R = 1.00$

$C_R = 1.00$

20. Determine the lifetime by using Table 9-12 OR given

Problem given: lifetime = 1000 hours



TABLE 9-12 Recommended Design Life

| Application | Design life (h) |
|---|---|
| Domestic appliances | 1000–2000 |
| Aircraft engines | 1000–4000 |
| Automotive | 1000–5000 |
| Agricultural equipment | 3000–6000 |
| Elevators, industrial fans, multipurpose gearing | 8000–15 000 |
| Electric motors, industrial blowers, general industrial machinery | 20 000–30 000 |
| Pumps and compressors | 40 000–60 000 |
| Critical equipment in continuous 24 h operation | 100 000–200 000 |

Source: Eugene A. Avallone and Theodore Baumeister III, eds. *Marks' Standard Handbook for Mechanical Engineers*, 9th ed. (New York: McGraw-Hill, 1987).

21, 22. Find the number of loading cycles ($N_{CP}$ and $N_{CG}$)

$N_{CP} = 60(\text{lifetime})(q)n_P = 60(1000)(300)(1) = 1.8 \times 10^7$ cycles

$N_{CG} = 60(\text{lifetime})(q)n_{G,a} = 60(1000)(100)(1) = 6 \times 10^6$ cycles

* Note: $q$ = # of load applications per revolution ($q = 1$)

$\therefore N_{CP} = 1.8 \times 10^7$ cycles, $N_{CG} = 6 \times 10^6$ cycles

22. Find the stress cycle factor ($K_L$) → Figure 10-16

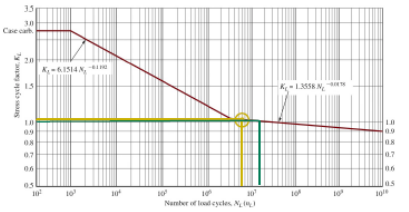



FIGURE 10-16 Stress cycle factor for bending strength, $K_L$ (carburized case-hardened steel bevel gears) (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Gear: $N_{LG} = 6 \times 10^6$ cycles YELLOW

$K_L = 1.3558\,N_L^{-0.0178}$

$K_L = 1.3558(6 \times 10^6)^{-0.0178}$

$K_{L,G} = 1.02694$

Pinion: $N_{LP} = 1.8 \times 10^7$ cycles GREEN

$K_L = 1.3558\,N_L^{-0.0178}$

$K_L = 1.3558(1.8 \times 10^7)^{-0.0178}$

$K_{L,P} = 1.00705$

23. Calculate Bending Strength Number ($S_{tP}$, $S_{tG}$)

Recall: $U_t = 1140.44$ lbf, $P_d = 10$ teeth/in, $K_o = 2.0$, $K_m = 1.25163$, $K_v = 1.07113$

$F = 1.25$, $J_G = 0.282$, $J_P = 0.274$

$$S_{tP} = \frac{U_t(P_d)K_o(K_R)(K_m)(K_v)}{(F)(J_P)} = \frac{(840)(10)(2.0)(0.5)(1.25163)(1.07113)}{(1.25)(0.274)}$$

$S_{tP} = 28968$ psi

$$S_{tG} = \frac{U_t(P_d)K_o(K_R)(K_m)(K_v)}{(F)(J_G)} = \frac{(840)(10)(2.0)(0.5)(1.25163)(1.07113)}{(1.25)(0.282)}$$

$S_{tG} = 37244.6$ psi

24. Find the maximum allowable strength ($S_{at}$)

$$S_{at,P} = \frac{S_{t,P}(SF)(K_R)}{K_{L,P}} = \frac{28968(1.00)(1.00)}{1.02694} = 28208.1$$

$S_{at,P} = 28208.1$ psi

$$S_{at,G} = \frac{S_{t,G}(SF)(K_R)}{K_{L,G}} = \frac{37244.6(1.00)(1.00)}{1.00705} = 36983.9$$

$S_{at,G} = 36983.9$ psi

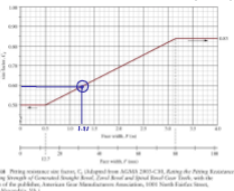
25. Pick material ⟶ steel

26. Find Elastic coefficient ($C_p$) ⟶ Table 9-7



For steel, $C_p = 2300$

27. Find size factor for contact stress ($C_s$) ⟶ Figure 10-18 or equation



@ F = 1.25 in

$C_s = 0.125F + 0.4375$

$= 0.125(1.25) + 0.4375$

$C_s = 0.59375$

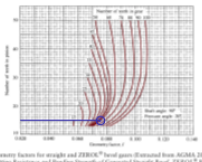
28. Find the crowning factor for pitting ($C_{xc}$)

$C_{xc} = 1.5$ for properly crowned teeth ⟵ usually pick $C_{xc} = 1.5$

$C_{xc} = 2.0$ or larger for non-crowned teeth

$C_{xc} = 1.5$

29. Find Geometry factor for Pitting resistance ($I$) ⟶ Figure 10-19



For $N_G = 45$, $H_P = 15$

$I = 0.075$

30. Find the stress cycle factor for contact stress ($C_L$) ⟶ Figure 10-20

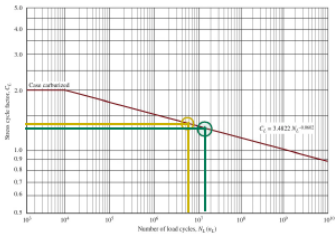


FIGURE 10-20 Stress cycle factor for pitting resistance, $C_L$ (carburized case-hardened steel bevel gears) (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Gear: $N_{LG} = 6 \times 10^8$ cycles YELLOW

$C_L = 3.4822\, N_L^{-0.0602}$

$C_{L,G} = 3.4822(6\times10^8)^{-0.0602}$

$C_{L,G} = 1.36085$

Pinion: $N_{L,P} = 1.8 \times 10^9$ cycles GREEN

$C_L = 3.4822\, N_L^{-0.0602}$

$C_{L,P} = 3.4822(1.8\times10^9)^{-0.0602}$

$C_{L,P} = 1.27376$

31. Find Pitting contact stress number ($S_c$)

Recall: $W_t = 1140.44$ lbf, $P_d = 10$ teeth/in, $K_o = 2.0$, $K_R = 0.5$, $K_m = 1.25563$, $K_v = 1.03119$

$F = 1.25$, $C_p = 10$, $C_s = 0.59375$, $C_{xc} = 1.5$, $I = 0.075$, $C_{L,G} = 1.36085$, $C_{L,P} = 1.27376$

$$S_c = C_p \sqrt{\frac{(W_t)(K_o)(K_m)(K_v)(C_s)(C_{xc})}{(F)(C_{D_P})(I)}}$$

$$S_c = (2300)\sqrt{\frac{(1140)(1.5)(1.25563)(0.59375)(1.5)}{(1.25)(1.5)(0.075)}}$$

$S_c = 238091$ psi

32. Find the Allowable Pitting Contact Stress ($S_{ac,P}$ & $S_{ac,G}$)

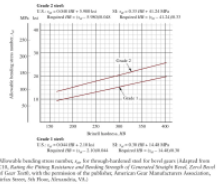
$$S_{ac,P} = \frac{(SF)(CR)}{C_{L,P}} S_c = \frac{(1.0)(1.0)}{1.27376}(238091) = 186920 \text{ psi}$$

$S_{ac,P} = 208710$ psi

$$S_{ac,G} = \frac{(SF)(CR)}{C_{L,G}} S_c = \frac{(1.0)(1.0)}{1.36085}(238091) = 174958 \text{ psi}$$

$S_{ac,G} = 195353$ psi

33. Calculate the required HB using Figure 10-17 (bending) & 10-21 (Pitting contact)



Grade 1 steel:

⟶ Gear: $HB = \frac{\frac{S_{at,P}}{1000} - 2.10}{0.044} = \frac{\frac{372448.6}{1000} - 2.10}{0.044}$

$HB_{G,\text{grade 1}} = 798.741$

Pinion: $HB = \frac{\frac{S_{at,P}}{1000} - 2.10}{0.044} = \frac{\frac{289681}{1000} - 2.10}{0.044}$

$HB_{P,\text{grade 1}} = 610.636$

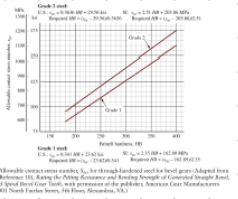
Grade 2 steel:

⟶ Gear: $HB_G = \frac{S_{at} - 5.980}{0.048} = \frac{\frac{372448.6}{1000} - 5.98}{0.048}$

$HB_{G,\text{grade 2}} = 651.346$

Pinion: $HB_P = \frac{S_{at} - 5.980}{0.048} = \frac{\frac{289681}{1000} - 5.98}{0.048}$

$HB_{P,\text{grade 2}} = 478.917$



Grade 1 steel ⟶ Gear: $HB_G = \frac{S_{ac} - 23.62}{0.341} = \frac{\frac{174958}{1000} - 23.62}{0.341} = 443.806$

$HB_{G,\text{grade 1}} = 443.806$

⟶ Pinion: $HB_P = \frac{S_{ac} - 23.62}{0.341} = \frac{\frac{186920}{1000} - 23.62}{0.341} = 478.856$

$HB_{P,\text{grade 1}} = 478.856$

Grade 2 steel ⟶ Gear: $HB_G = \frac{S_{ac} - 29.56}{0.3636} = \frac{\frac{174958}{1000} - 29.56}{0.3636} = 399.884$

$HB_{G,\text{grade 2}} = 399.884$

Pinion: $HB_P = \frac{S_{ac} - 29.56}{0.3636} = \frac{\frac{186920}{1000} - 29.56}{0.3636} = 432.983$

$HB_{P,\text{grade 2}} = 432.783$

Pick $HB_{P,\text{grade 1}} = 610.636$

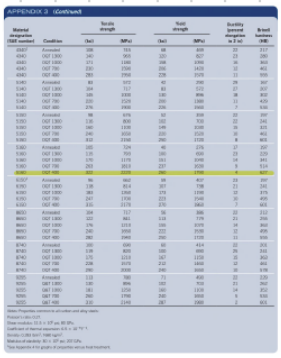
* Note this is because the question said to ONLY look at Pinion

34. Use Appendix 3, 4, 5 to pick the suitable material

→ Pick HB = 542.786

→ HB from table > 542.786

Appendix 3



Specific Heat treatment

→ Appendix 4

For material ②

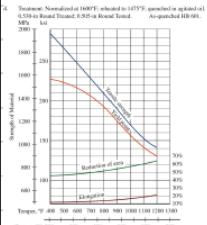




FIGURE A4-5 Properties of heat-treated SAE 4340, oil-quenched and tempered

→ But Appendix doesn't contain Heat treatment for the selected material

From online source



reference

https://www.sdknives.co.za/product/sae5160-high-carbon-alloy-steel-6mm-x-60mm-x-1m/#:~:text=HEAT%20TREATMENT%20OF%205160&text=843°C%2C%20Hold%20for,air%20to%20black%20and%20magnetic.

Gear and Pinion have same material

① HB627: SAE 5160 OGT 400 at $\sigma_{tensile}$ = 322 ksi, $\sigma_{yield}$ = 260 ksi, 4 % elongation

35. Calculate Final safety factor from a chosen material using Figure 10-17 & 10-21

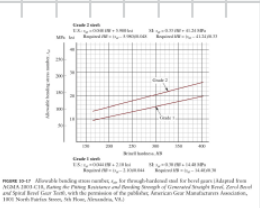


FIGURE 10-17 Allowable bending stress number, $\sigma_{at}$, for through-hardened steel for bevel gears (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

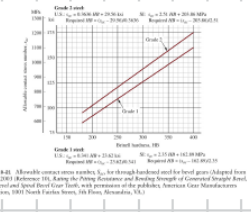


FIGURE 10-21 Allowable contact stress number, $\sigma_{ac}$, for through-hardened steel for bevel gears (Adapted from AGMA 2003 (Reference 10), *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Grade 1

$S_{at}$ = 0.044 HB + 2.10

= 0.044 (627) + 2.10

$S_{at}$ = 29.688 ksi

Bending

$$S_{Fp} = \frac{S_{at,P}}{S_{tP}} \frac{K_{L,P}}{K_R} = \frac{(29.688\times10^3)(1.02694)}{(28968)(1.00)} = 1.05246$$

$S_{Fp}$ = 1.05246

$$S_{Fc} = \frac{S_{at,G}}{S_{tG}} \frac{K_{L,G}}{K_R} = \frac{(29.688\times10^3)(1.08705)}{(32249.6)(1.00)} = 0.802748$$

$S_{Fc}$ = 0.8027

Grade 1

$S_{ac}$ = 0.341 HB + 23.62

= 0.341 (627) + 23.62

$S_{ac}$ = 237.427 ksi

Pitting contact

$$S_{Hp} = \frac{S_{ac,P}}{S_{c,P}} \frac{C_{L,P}}{K_R} = \frac{(237.427\times10^3)(1.27776)}{238091\ (1.00)} = 1.2742$$

adjust $S_{Hp} = S_{Hp}^2 = (1.2746)^2 = 1.62358$

adjusted $S_{Hp}$ = 1.62358

$$S_{Hc} = \frac{S_{ac,G}}{S_{c,G}} \frac{C_{L,G}}{K_R} = \frac{(237.427\times10^3)(1.36081)}{238091\ (1.00)} = 1.35705$$

$S_{HG} = S_{HG}^2 = (1.35705)^2 = 1.8416$

adjusted $S_{HG}$ = 1.8416

∴ 1. Pitch diameter is → $D_P$ = 1.5 in with cone angle $\delta$ = 18.4°, $D_G$ = 4.5 in with cone angle $\Gamma$ = 71.6°

2. $W_t$ = 540 lbf → $W_{r,G}$ = 96.6568 lbf and $W_{x,G}$ = 290.054 lbf, $W_{r,P}$ = 290.054 lbf and $W_{x,P}$ = 96.6568 lbf

3. Bending stress number of pinion ($S_{tp}$) = 28968 psi

Pitting contact number of pinion ($S_{cp}$) = 238091 psi

4. Selected material is HB627: SAE 5160 OGT 400 at $\sigma_{tensile}$ = 322 ksi, $\sigma_{yield}$ = 260 ksi, 4 % elongation with normalized temperature of 843°C, hold for 10 minutes then cool in still air til black and magnetic

5. Final factor of safety of pinion → Bending Factor of safety ($S_{Fp}$) = 1.05246

Pitting Contact factor of safety ($S_{HG}$) = 1.8416